SOICT

HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 1.1: Introduction

Dam Quang Tuan

Lecture Objectives

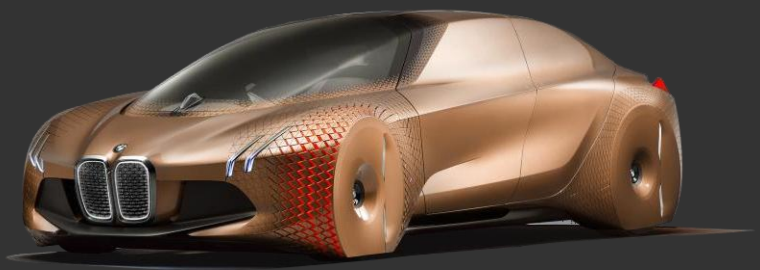
- What is Multimodal?
  - Research-oriented definition
  - Dimensions of modality heterogeneity
  - Modality connections and interactions
- Core technical and conceptual challenges
  - Representation, alignment, reasoning, generation, transference and quantification
- Course schedule

What is Multimodal?

Multimodal AI Technologies

Robots

Personal Vehicles



Ubiquitous

Mobile

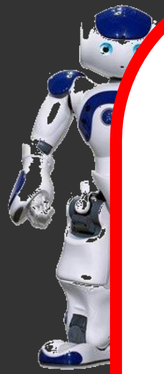
Wearable

Online

YouTube

Multimodal AI Technologies

Robots

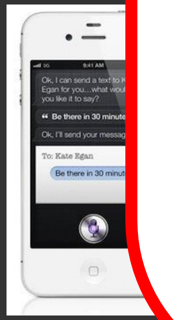


Personal Vehicles

Ubiquitous

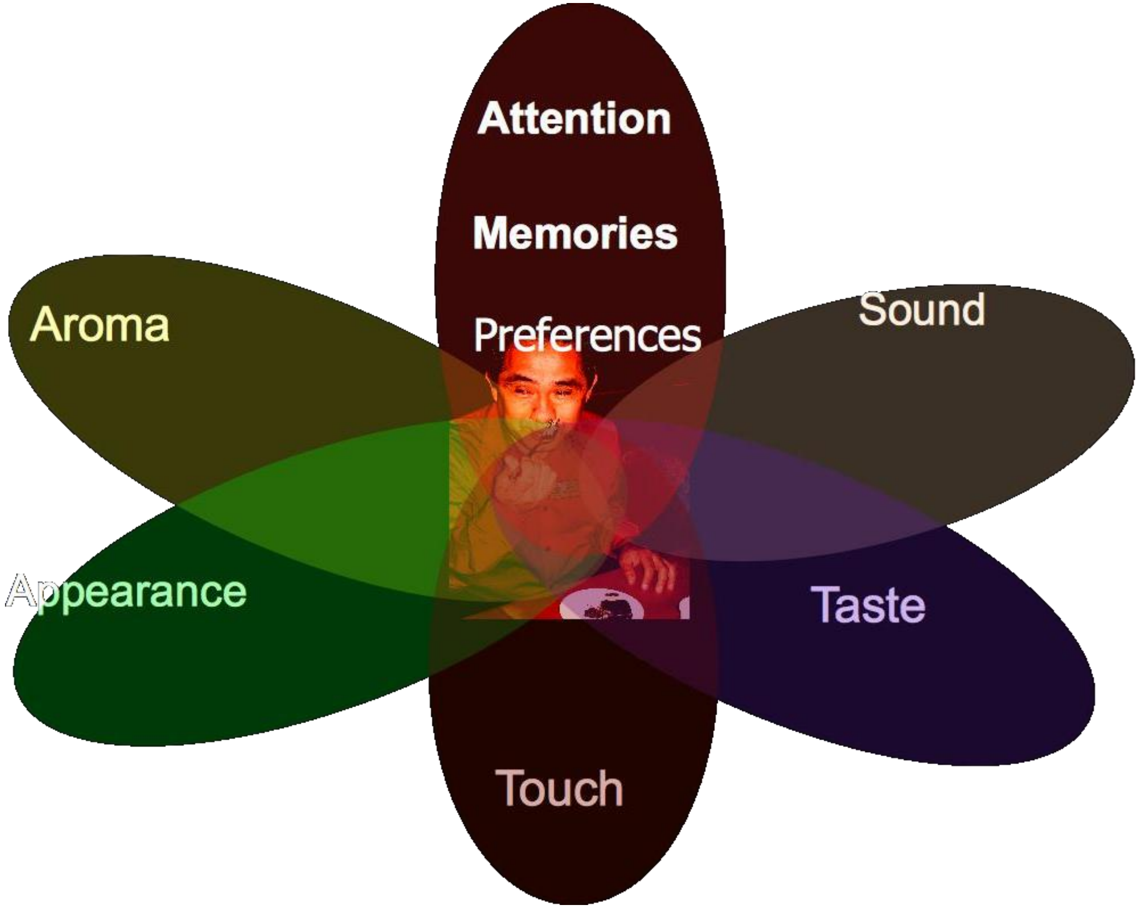
Video Conferencing

M



Wearable

What is Multimodal?



Sensory Modalities

Multimodal Behaviors and Signals

Language
- **Lexicon**
  - Words
- **Syntax**
  - Part-of-speech
  - Dependencies
- **Pragmatics**
  - Discourse acts

Acoustic
- **Prosody**
  - Intonation
  - Voice quality
- **Vocal expressions**
  - Laughter, moans

Visual
- **Gestures**
  - Head gestures
  - Eye gestures
  - Arm gestures
- **Body language**
  - Body posture
  - Proxemics
- **Eye contact**
  - Head gaze
  - Eye gaze
- **Facial expressions**
  - FACS action units
  - Smile, frowning

Touch
- **Haptics**
- **Motion**

Physiological
- **Skin conductance**
- **Electrocardiogram**

Mobile
- **GPS location**
- **Accelerometer**
- **Light sensors**

Multimodal Behaviors and Signals

High-Modality Multimodal Transformer: Quantifying Modality &Interaction Heterogeneity for High-Modality Representation Learning Liang et al 2023

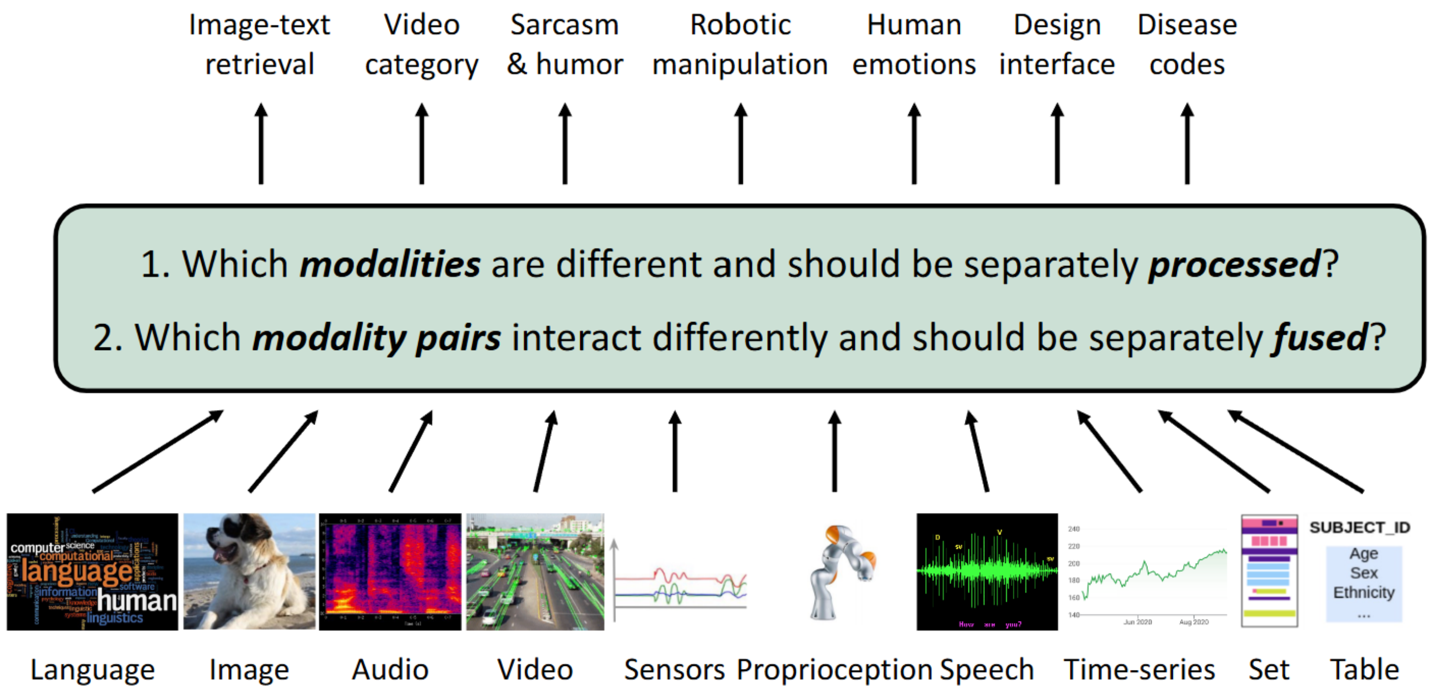
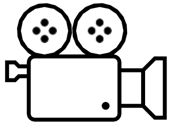


Figure 1: **Heterogeneity quantification**: Efficiently learning from many modalities requires measuring (1) *modality heterogeneity*: which modalities are different and should be separately processed, and (2) *interaction heterogeneity*: which modality pairs interact differently and should be separately fused. HighMMT uses these measurements to dynamically group parameters balancing performance and efficiency.

What is a Modality?

Modality

Modality refers to the way in which something expressed or perceived.



from a sensor

Raw Modalities
(closest from sensor)

Abstract Modalities
(farthest from sensor)

Examples:

Speech signal	Language	Sentiment intensity
Image	Detected objects	Object categories

What is Multimodal?

A dictionary definition…

Multimodal: with multiple modalities

A research-oriented definition…

***Multimodal* is the science of**

heterogeneous and interconnected data

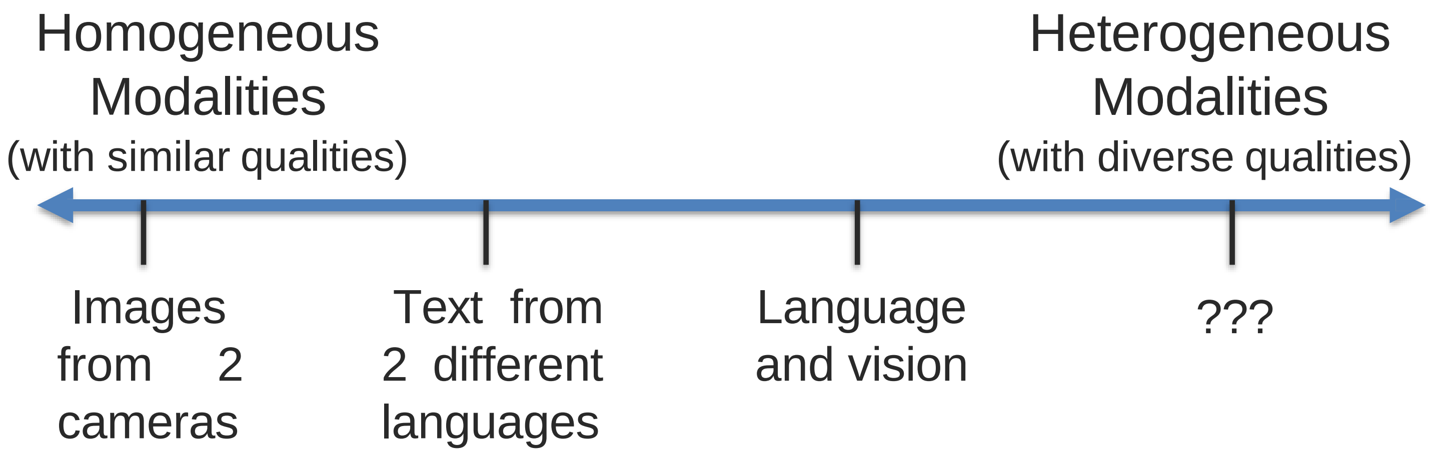
Heterogeneous Modalities

Information present in different modalities will often show diverse qualities, structures and representations.

Modality A

Modality B

Examples:



Abstract modalities are more likely to be homogeneous

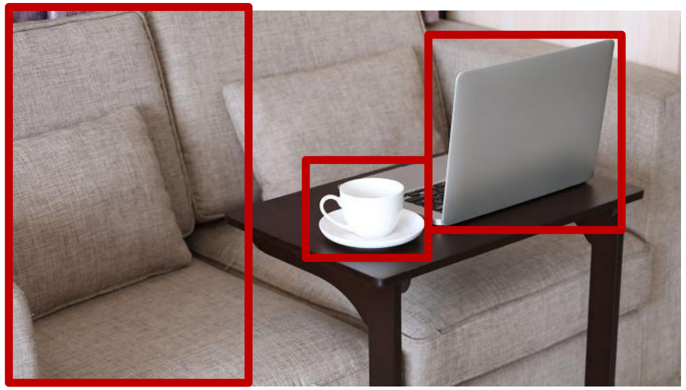
Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.

A teacup on the right of a laptop in a clean room.

Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.



*A **teacup** on the **right** of a **laptop** in a **clean room**.*

1 **Element representations:** discrete, continuous, granularity

{  ,  ,  }

{teacup, right, laptop, clean, room}

Dimensions of Heterogeneity

"Cross-Modal Discrete Representation Learning" by Alexander H. Liu et al

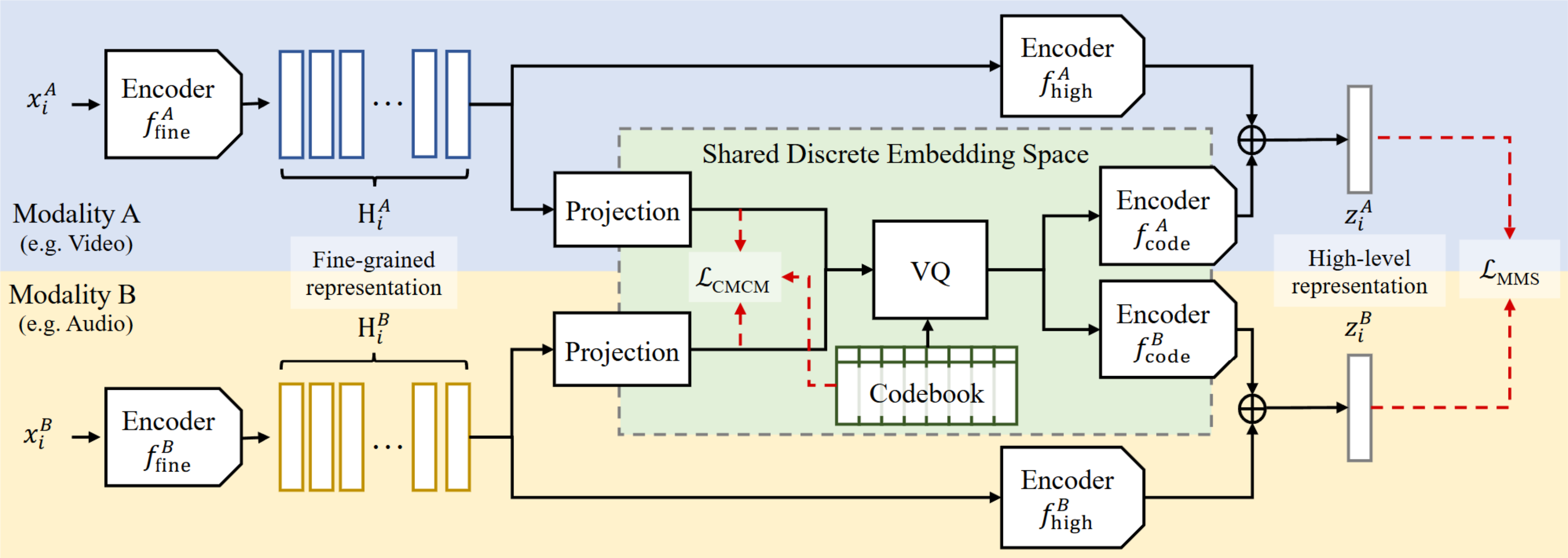
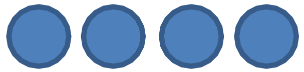


Figure 1: An overview of the proposed framework. The proposed shared discrete embedding space (green region, described in Section 2.2) is based on a cross-modal representation learning paradigm (blue and yellow regions, described in Section 2.1). The proposed Cross-Modal Code Matching $\mathcal{L}_{\text{CMCM}}$ objective is detailed in Section 2.3 and Figure 2.

Dimensions of Heterogeneity

2 **Element distributions:** density, frequency

objects per image



words per minute

•Different modalities have different **densities and frequencies** of information.

- Text: Words per minute.
- Vision: Objects per image.
- Audio: Frequencies of sound waves.

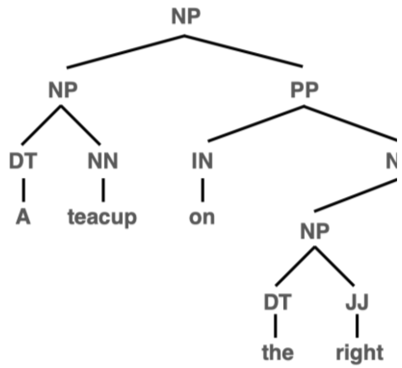
Example

•**Speech signals** are **dense** (continuous waveforms).

•**Text documents** are **sparse** (symbolic, limited vocabulary).

Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.



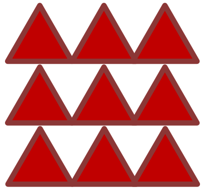
…

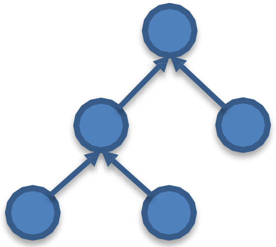
Latent
(implicit)

A teacup on the right …

Explicit
(observable)

3 **Structure:** temporal, spatial, hierarchical, latent, explicit





Dimensions of Heterogeneity

Structural Differences

•Data can be **temporal, spatial, hierarchical, latent, or explicit**:

- **Temporal**: Speech signals evolve over time.
- **Spatial**: Objects in an image have spatial relationships.
- **Hierarchical**: Sentences follow syntax trees.
- **Latent**: Hidden features extracted by deep learning models.
- **Explicit**: Directly observable structures.

Example

•**Speech:** Time-series structure (temporal).

•**Image:** Object layout in a scene (spatial).

•**Text:** Syntactic dependency trees (hierarchical).

A teacup on the right …

Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.

A teacup on the right of a laptop in a clean room.

4 **Information:** abstraction, entropy

Dimensions of Heterogeneity

Information Abstraction & Entropy

•Some modalities provide **high-level abstract information** (e.g., text summarization), while others offer **raw detailed data** (e.g., image pixels).

•**Entropy** measures **uncertainty** in the data:

- Higher entropy = More variability (e.g., spoken dialogue).
- Lower entropy = More structured data (e.g., written text).

Example

•**A GPS location ("40.7128° N, 74.0060° W")** is **low-entropy** (specific).

•**A text description ("Somewhere in New York")** is **high-entropy** (ambiguous).

Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.

A teacup on the right of a laptop in a clean room.

5 **Noise:** uncertainty, signal-to-noise ratio, missing data

teacup ⟶ teacip

right ⟶ rihjt

Dimensions of Heterogeneity

Noise & Uncertainty

•Some modalities are **noisier** than others:

- **Speech signals** have background noise.
- **Text transcriptions** may have typos.
- **Camera images** may be blurry.

Example

•OCR (Optical Character Recognition) misreading "teacup" as "teacip".

•Speech-to-text systems misinterpreting "right" as "write".

5 **Noise:** uncertainty, signal-to-noise ratio, missing data

teacup ⟶ teacip

right ⟶ rihjt

Dimensions of Heterogeneity

Information present in different modalities will often show diverse qualities, structures, and representations.

A teacup on the right of a laptop in a clean room.

6 **Relevance:** task relevance, context dependence

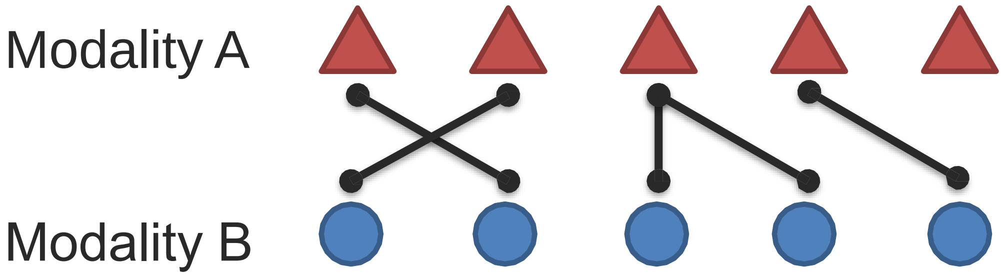
→ recreational
→ living room
→ right-handed

A teacup on the right of a laptop in a clean room. → workspace
→ study room

Interconnected Modalities

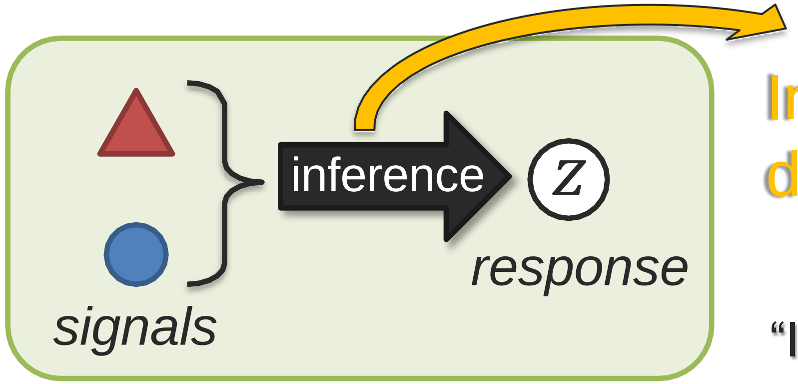
① **Modality connections**

Modalities are often related and share commonality



② **Modality interactions**

Modality elements often interact during inference



Interactions happen during inference!

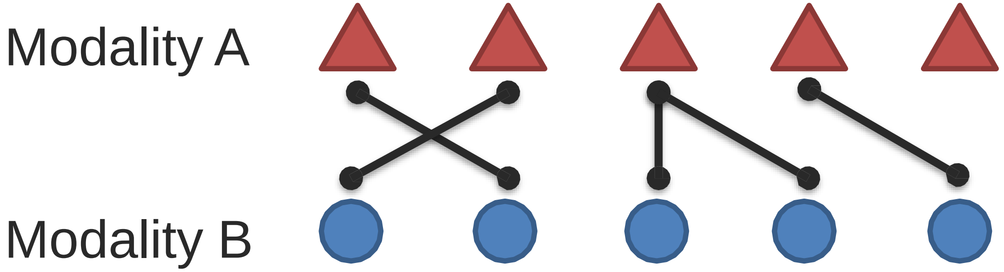
"Inference" examples:

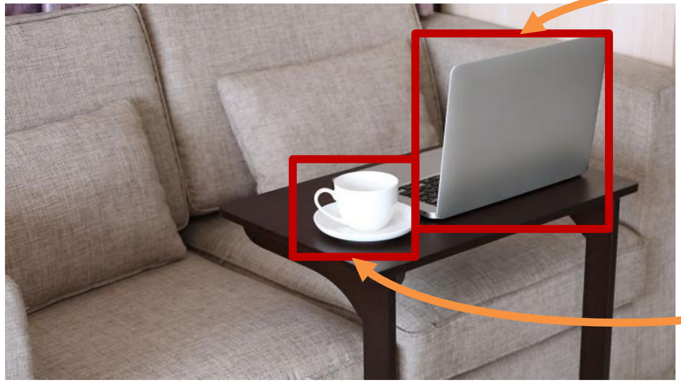
- Behavior perception
- Recognition task
- Modality translation

Interconnected Modalities

① Modality connections

Modalities are often related
and share commonality



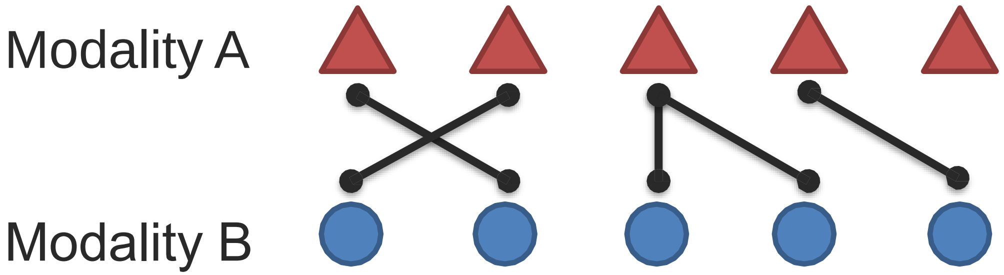


A ***teacup*** *on the right of a* ***laptop***
in a clean room.

Interconnected Modalities

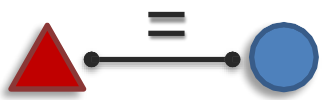
① Modality connections

Modalities are often related and share commonality



Statistical

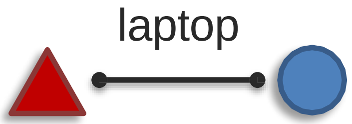
Association



e.g., correlation, co-occurrence

Semantic

Correspondence

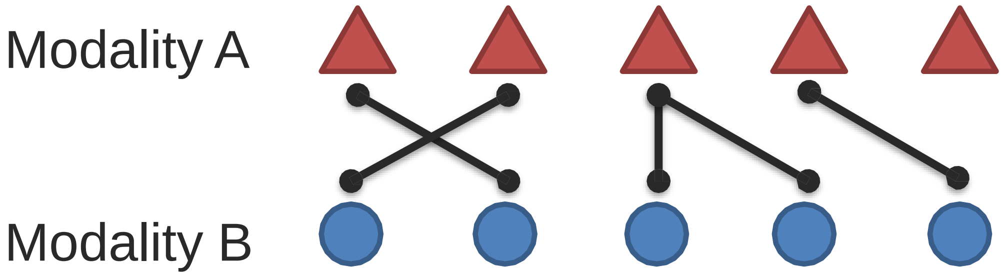


e.g., grounding

Interconnected Modalities

① Modality connections

Modalities are often related and share commonality

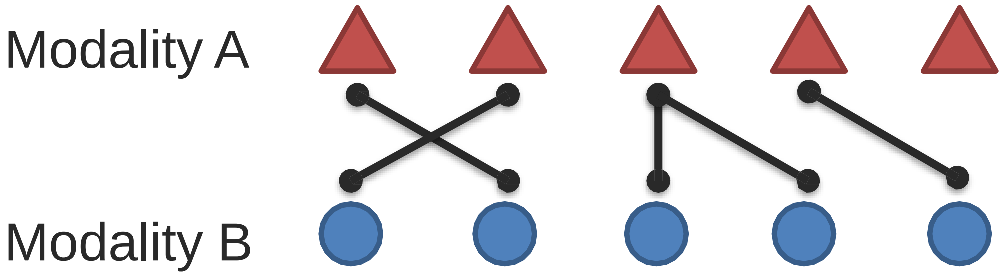


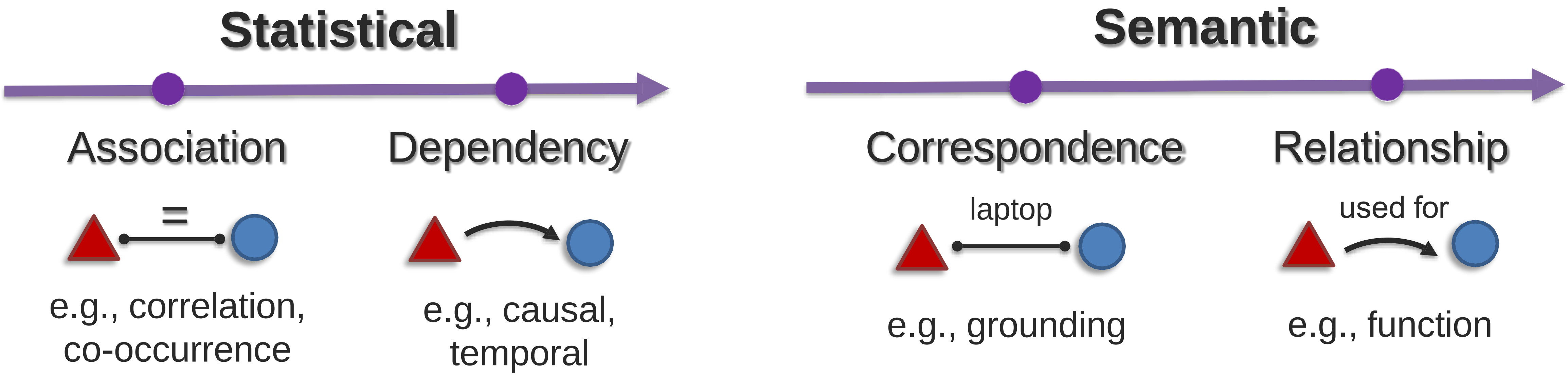
*A teacup on the right of a laptop in a **clean room**.*

Interconnected Modalities

① Modality connections

Modalities are often related and share commonality





Interconnected Modalities

② **Modality interactions**

Modality elements often interact during inference

signals } inference → ○ *response* → Types of interaction responses? (a taxonomy)

Is this indoors?

inference → *Yes!*

A teacup on the right of a laptop in a clean room.

inference → *Yes!*

Unimodal redundancy

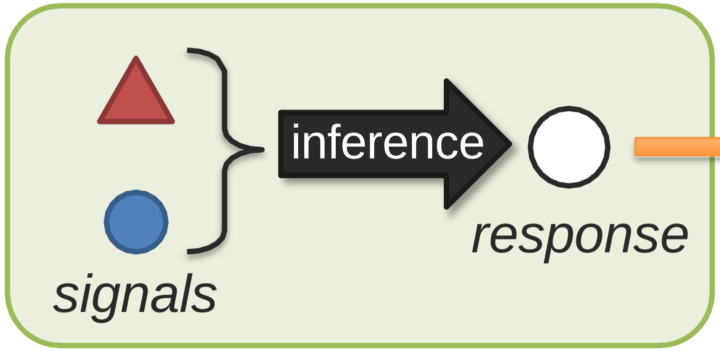
▲ → □

● → □

Interconnected Modalities

② Modality interactions

Modality elements often interact during inference



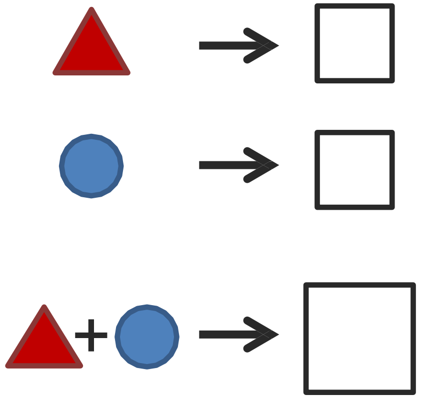
Types of interaction responses?

Is this indoors?

A teacup on the right of a laptop in a clean room.

inference **Yes!**

Unimodal redundancy

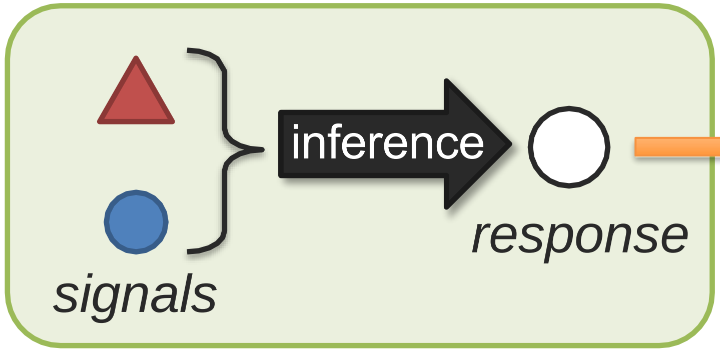


Multimodal enhancement

Interconnected Modalities

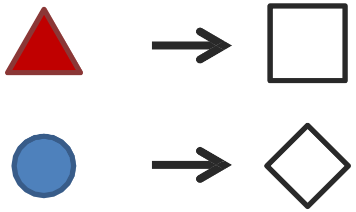
② Modality interactions

Modality elements often interact during inference



Types of interaction responses?

Unimodal
Non-redundancy



Is this a living room?

inference → *Yes!*

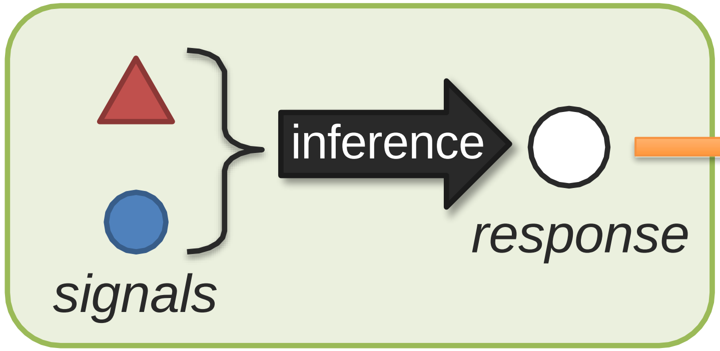
A teacup on the right of a laptop in a clean room.

inference → *No, probably study room.*

Interconnected Modalities

② Modality interactions

Modality elements often interact during inference



Types of interaction responses?

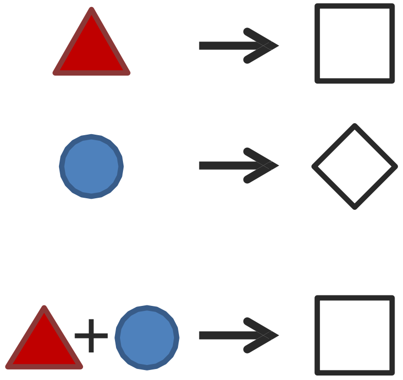
Is this a living room?

A teacup on the right of a laptop in a clean room.

inference

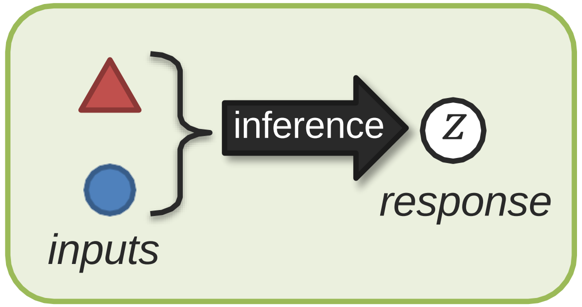
Yes!

Unimodal
Non-redundancy

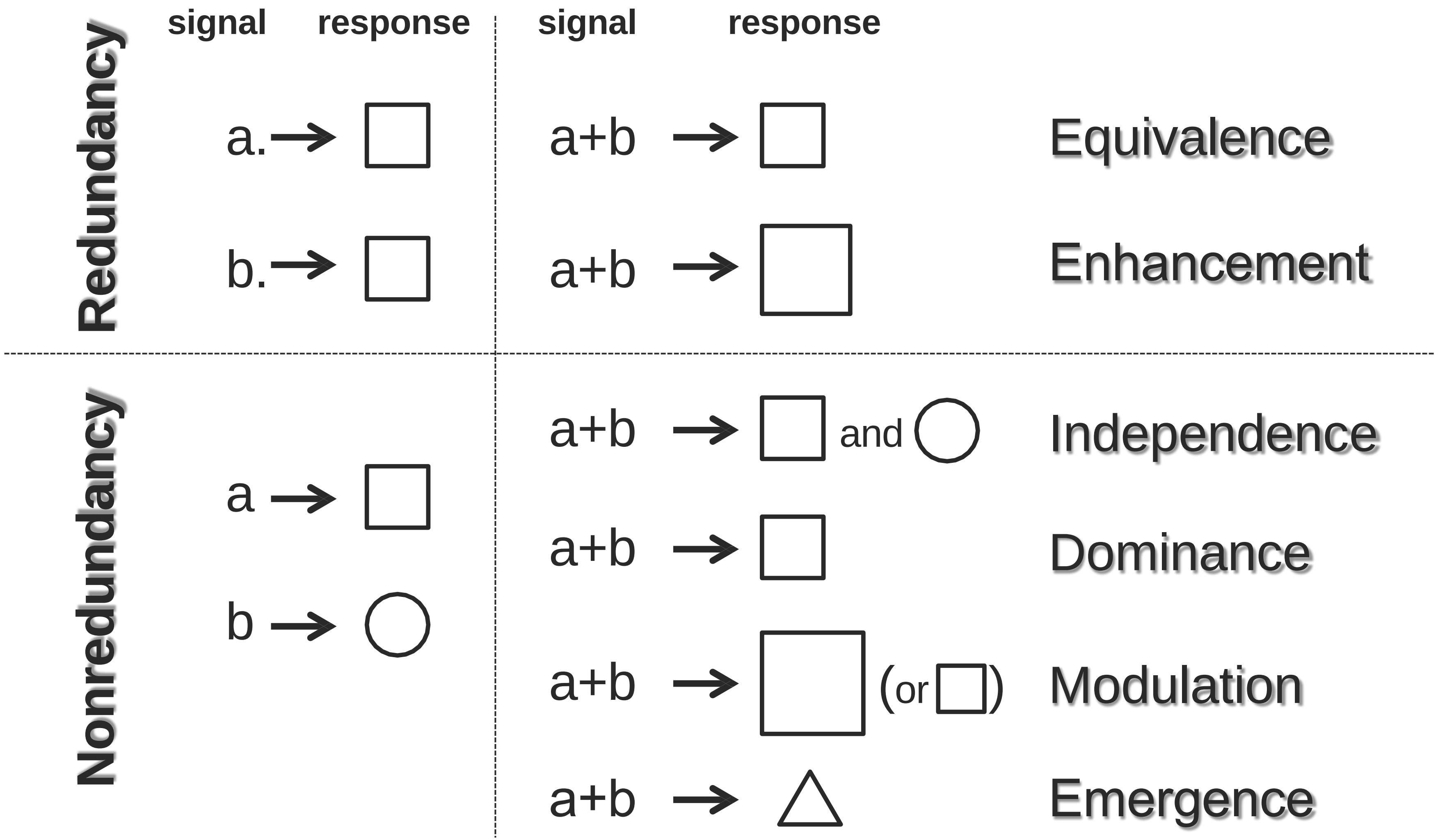


Multimodal
dominance

Taxonomy of Interaction Responses – A Behavioral Science View



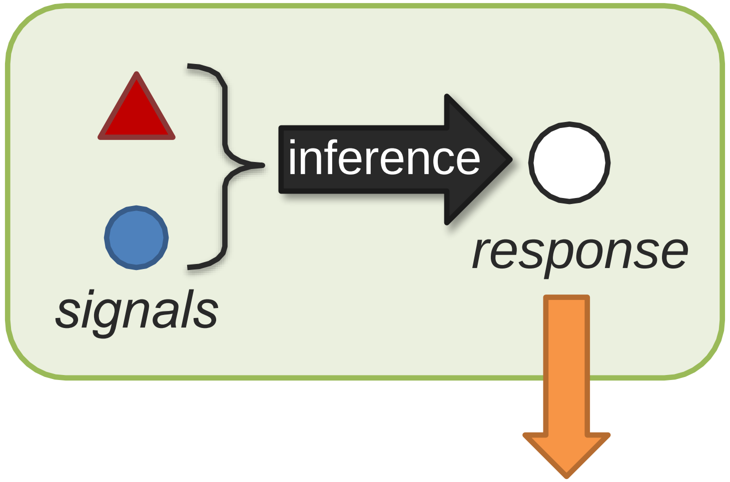
Multimodal Communication



Partan and Marler (2005). *Issues in the classification of multimodal communication signals*. American Naturalist, 166(2)

Dimensions of Modality Interactions

What are the dimensions for **digitally-represented** modalities?

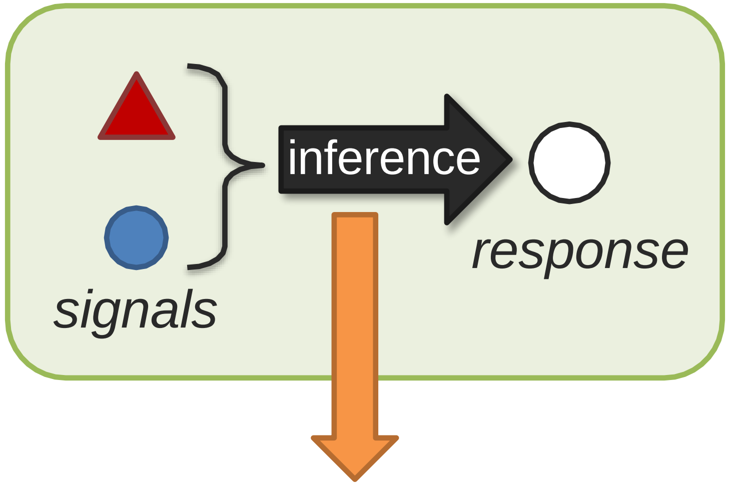


1 Interaction Responses:

- Redundancy
- Non-redundancy
- Dominance
- Emergence…

Dimensions of Modality Interactions

What are the dimensions for **digitally-represented** modalities?

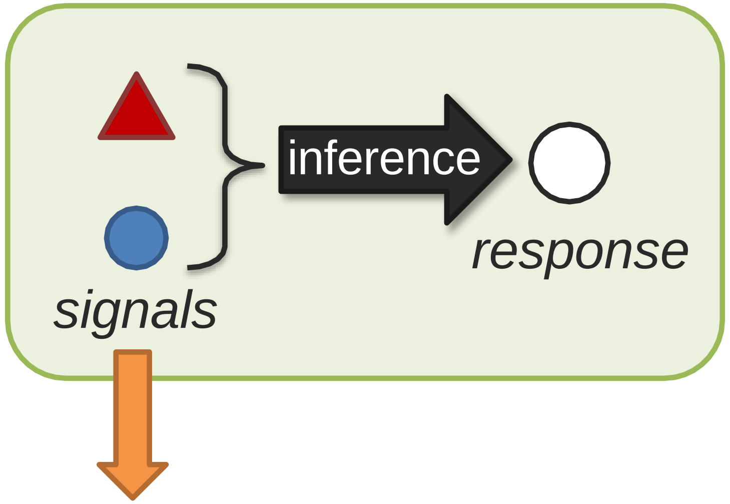


2 Interaction Mechanics:

- Additive
- multiplicative
- Nonlinear
- Causal,
- Logical, …

Dimensions of Modality Interactions

What are the dimensions for **digitally-represented** modalities?

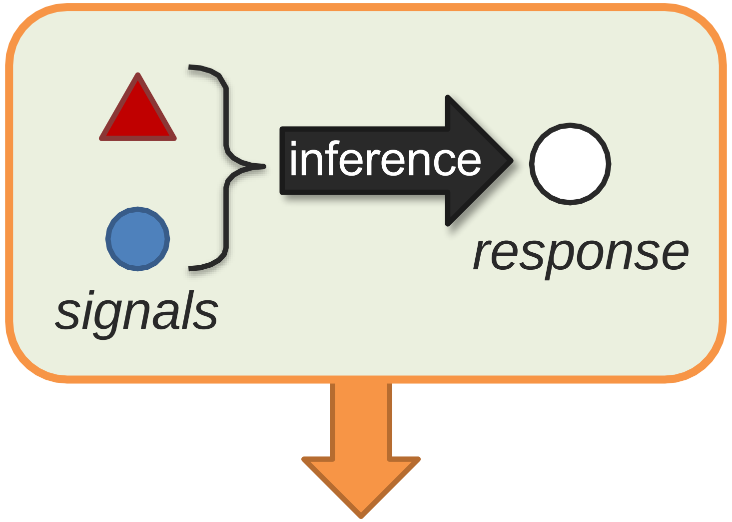


3 Input modalities:

- Unimodal
- Bimodal
- Trimodal
- High-modal, ...

Dimensions of Modality Interactions

What are the dimensions for **digitally-represented** modalities?



4 Context:

- Structure context
- Task relevance
- Context dependence
- High-modal, ...

What is Multimodal?

Multimodal **is the science of**

heterogeneous and interconnected data ☺

Multimodal Machine Learning

What is Multimodal Machine Learning?

Multimodal Machine Learning (ML) is the study of computer algorithms that learn and improve through the use and experience of data from multiple modalities

Multimodal Artificial Intelligence (AI) studies computer agents able to demonstrate intelligence capabilities such as understanding, reasoning and planning, through multimodal experiences, and data

Multimodal AI is a superset of **Multimodal ML**

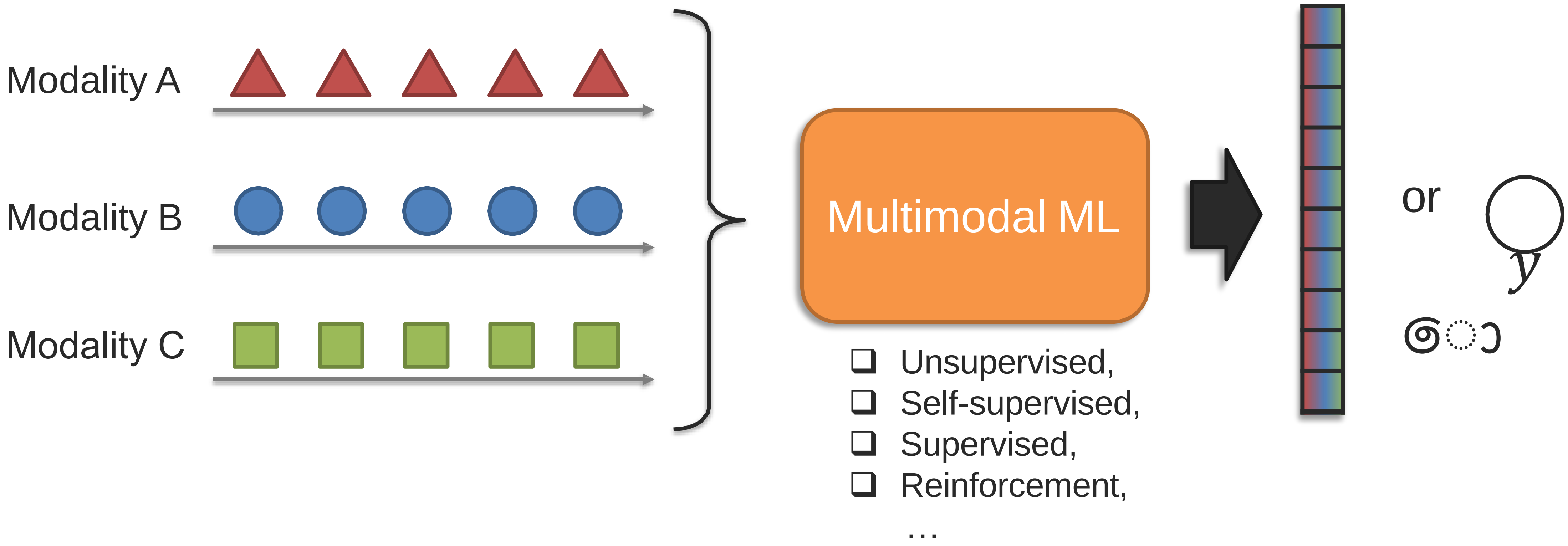
Multimodal Machine Learning

Language I really like this tutorial

Vision

Acoustic

Multimodal Machine Learning



*What are the **core multimodal technical challenges**, understudied in conventional machine learning?*

Multimodal Technical Challenges – Surveys, Tutorials and Courses

2016

Multimodal Machine Learning: A Survey and Taxonomy

Tadas Baltrusaitis, Chaitanya Ahuja and Louis-Philippe Morency

(Arxiv 2017, IEEE TPAMI journal, February 2019)

https://arxiv.org/abs/1705.09406

Tutorials: CVPR 2016, ACL 2016, ICMI 2016, …

Graduate-level courses:

Multimodal Machine learning (11th edition)

https://cmu-multicomp-lab.github.io/mmml-course/fall2020/

Advanced Topics in Multimodal Machine learning

https://cmu-multicomp-lab.github.io/adv-mmml-course/spring2022/

2022

Fundamentals of Multimodal ML: A Taxonomy & Open Challenges

Paul Liang, Amir Zadeh and Louis-Philippe Morency

- ☑ 6 core challenges
- ☑ 50+ taxonomic classes
- ☑ 600+ referenced papers

Tutorials: CVPR 2022, NAACL 2022, …

Updated graduate-level course:

Multimodal Machine learning (12th edition)

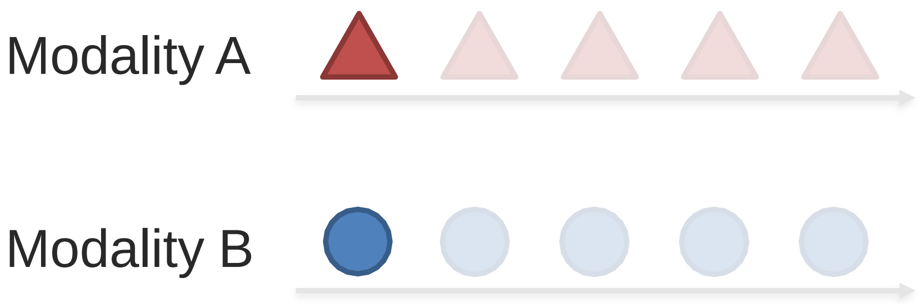
Fall 2022 semester

Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

This is a **core challenge** in multimodal learning because different modalities (text, images, audio, etc.) have **diverse structures** and need to be unified in a meaningful way.

Individual elements:



It can be seen as a "local" representation

or

representation using holistic features

Challenge 1: Representation

Example 1: Vision-Language Representation (CLIP)

•**Task**: Match images with text descriptions.

•**Problem**: Text and images have different feature spaces.

•**Solution**: CLIP (Contrastive Language–Image Pretraining) learns a **shared latent space** where related text and images are close together.

Learning Transferable Visual Models From Natural Language Supervision Radford 2021

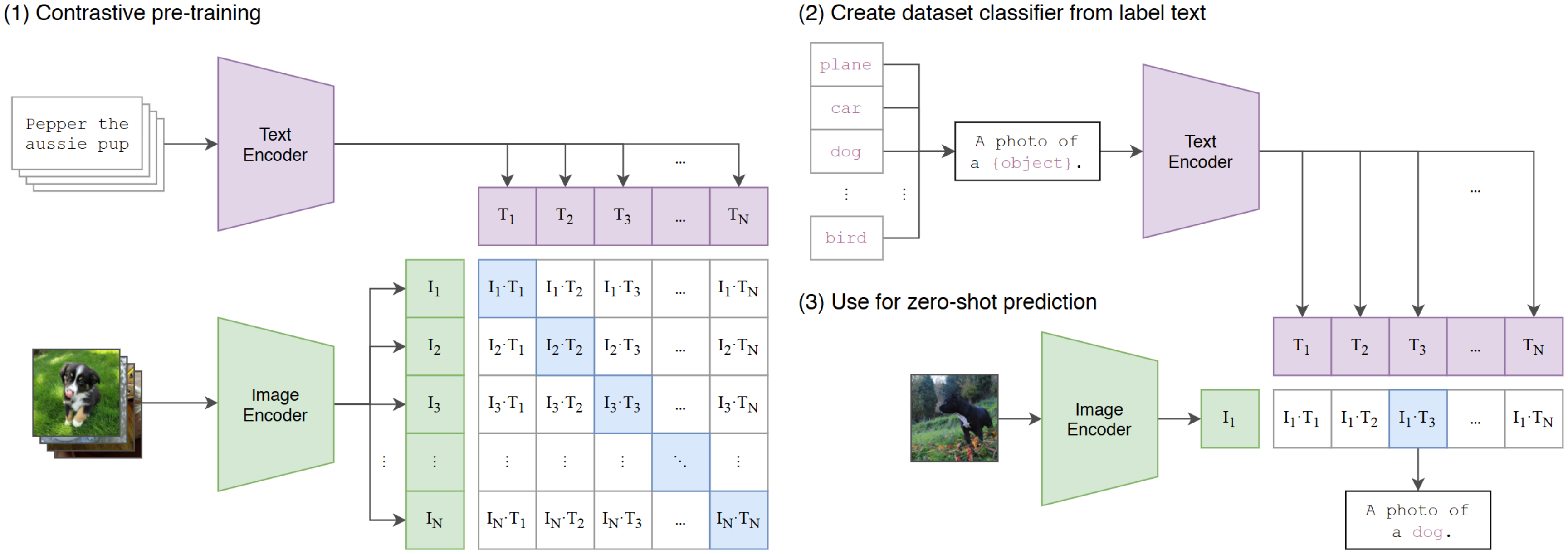


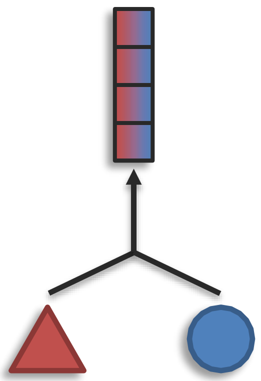
Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

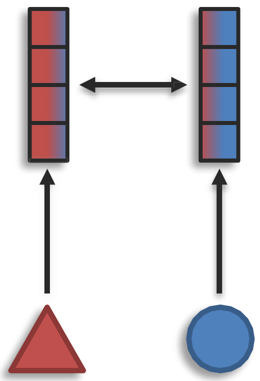
Sub-challenges:

Fusion



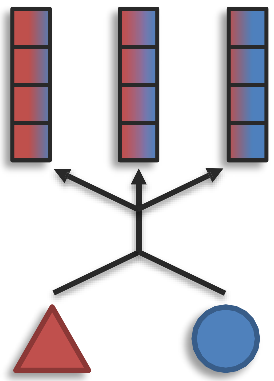
modalities **>** # representations

Coordination



modalities **=** # representations

Fission



modalities **<** # representations

Challenge 1: Representation

Dynamic Multimodal Fusion 2023

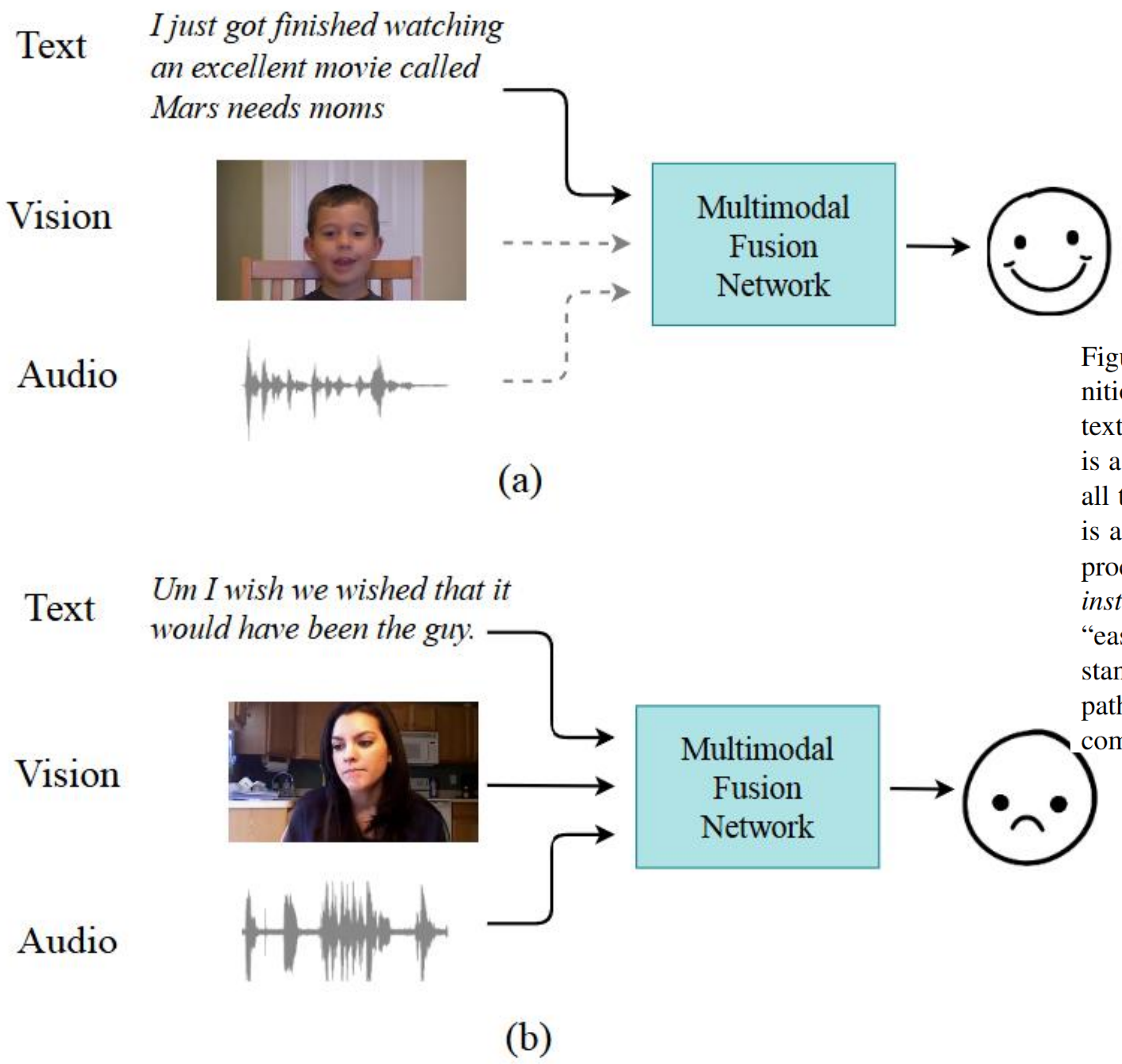


Figure 1. Two examples in CMU-MOSEI [51] for emotion recognition. Figure (a) shows an "easy" multimodal instance as using textual information is sufficient to predict emotions correctly (this is a positive emotion). Figure (b) shows a "hard" example where all three modalities are required to make correct predictions (this is a negative emotion). While static multimodal fusion networks process "hard" and "easy" inputs identically, we propose *dynamic instance-wise inference* that can achieve computational savings for "easy" examples and preserve representation power for "hard" instances. For (a), DynMM only activates the text path and skips paths corresponding to the other two modalities, thus leading to computational efficiency.

Challenge 1: Representation

Mirror U-Net: Marrying Multimodal Fission with Multi-task Learningfor Semantic Segmentation in Medical Imaging, Marinov 2023

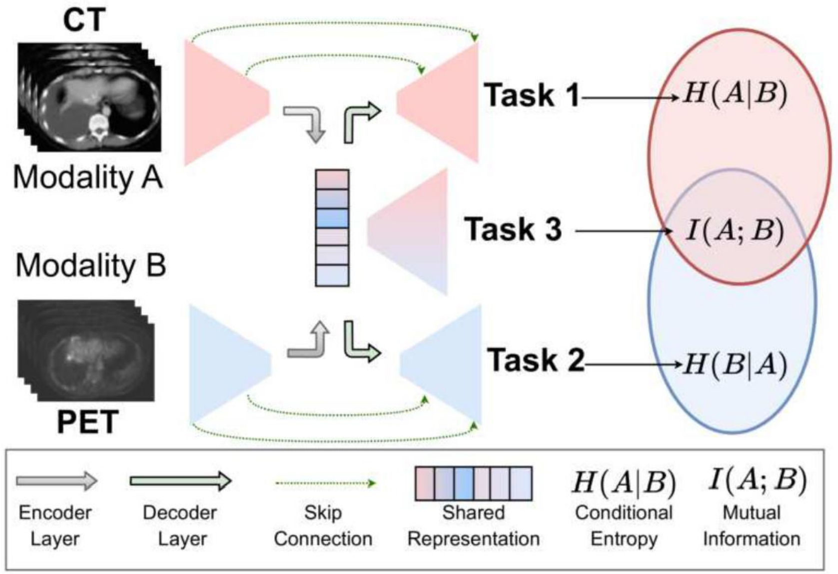


Figure 1. Mirror U-Net combines multimodal fission [26] and multi-task learning. We obtain a shared representation for both modalities and feed it into modality-specific decoders, each optimized for a tailored task to learn useful features from its modality. Tasks 1 and 2 use modality-specific features via the skip connections but are also conditioned by the other modality via the shared representation, hence focusing on the conditional entropy between the modalities. Task 3, on the other hand, only processes the shared representation, focusing on the mutual information.

Challenge 2: Alignment

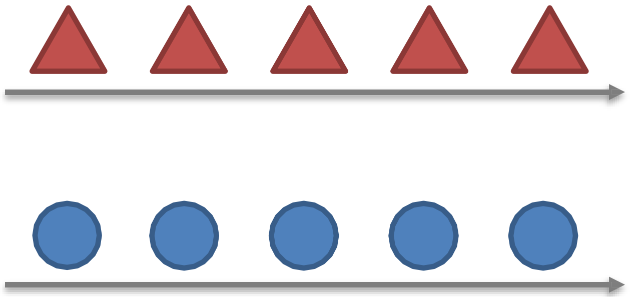
Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

Most modalities have internal structure with multiple elements

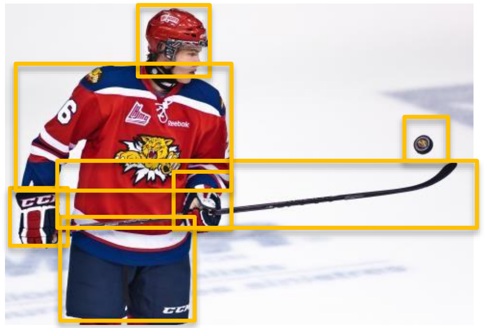
Elements with temporal structure:

Modality A

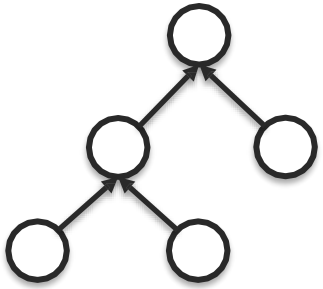
Modality B



Other structured examples:



Spatial



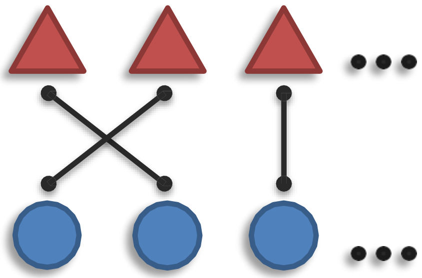
Hierarchical

Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

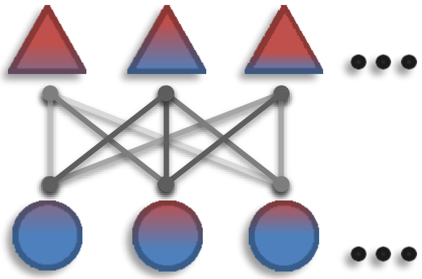
Sub-challenges:

Connections



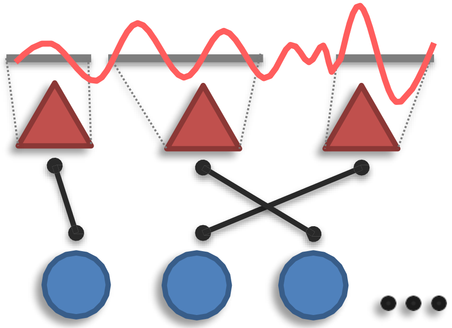
Explicit alignment
(e.g., grounding)

Aligned Representation



Alignment + representation
(aka, contextualized representation)

Elements



Segmentation of individual elements

Challenge 2: Alignment

Aligned Representation (aka, Contextualized Representation)

Definition:

•Learning a **joint representation** that **preserves relationships** across modalities.
•Instead of direct mapping, a **shared space** is used for interaction.

Example 1: Contrastive Learning for Vision-Language Models (CLIP)

•**Task**: Match images with their textual descriptions.
•**Challenge**: Text and images exist in **separate feature spaces**.
•**Solution**: Contrastive models (e.g., **CLIP, ALIGN**) learn a **shared latent space** where images and their matching captions are closer together.
•**Concrete Example**:

- Image 1: 🌄 **"A sunset over the ocean"** → 🟢 Close to caption: **"A beautiful sunset with waves."**
- Image 2: 🏙️ **"A city skyline at night"** → ❌ **Not close to sunset description**.

Challenge 2: Alignment

Aligned Representation (aka, Contextualized Representation)

Definition:

- Learning a **joint representation** that **preserves relationships** across modalities.
- Instead of direct mapping, a **shared space** is used for interaction.

Example 2: Audio-Visual Representation for Lip-Reading

- **Task**: Improve speech understanding by **incorporating visual cues (lip movements)**.
- **Challenge**: Speech and video signals have **different temporal characteristics**.
- **Solution**: **Multimodal transformers (e.g., AV-HuBERT, MM-ViT)** align **spoken words** with **lip movements** in a joint space.
- **Concrete Example**:
    - **Video**: Person says "Good morning" but **audio is noisy**.
    - **Aligned Representation**: The model infers missing words using **lip movement embeddings**.

Challenge 2: Alignment

Elements: Segmentation of Individual Elements

Definition:

•Breaking multimodal data into **smaller, meaningful elements** before processing.

•Essential for **structured** data like time-series, hierarchical, or sequential inputs.

Example 1: Speech Processing – Phoneme Segmentation

•**Task**: Split an **audio signal** into **phonemes** (smallest units of sound).

•**Challenge**: Words in speech **do not have clear boundaries**.

•**Solution**: Models like **Wav2Vec, HuBERT** segment speech into **phonemes** before transcribing.

•**Concrete Example**:

- **Audio Input**: "Elephant"
- **Segmented Elements**: ["Eh" - "luh" - "fuh" - "uhnt"]

Challenge 2: Alignment

Elements: Segmentation of Individual Elements

Definition:

•Breaking multimodal data into **smaller, meaningful elements** before processing.

•Essential for **structured** data like time-series, hierarchical, or sequential inputs.

Example 2: Video Action Recognition – Motion Segmentation

•**Task**: Identify **distinct actions** in a continuous video stream.

•**Challenge**: Actions **blend into each other** (e.g., running → jumping).

•**Solution**: **Temporal action segmentation models (e.g., SlowFast, TimeSformer)** split the video into **meaningful units**.

•**Concrete Example**:

- **Video Input**: "A person picks up a glass, drinks water, then puts it down."
- **Segmented Actions**: [pick-up] → [drink] → [put-down]

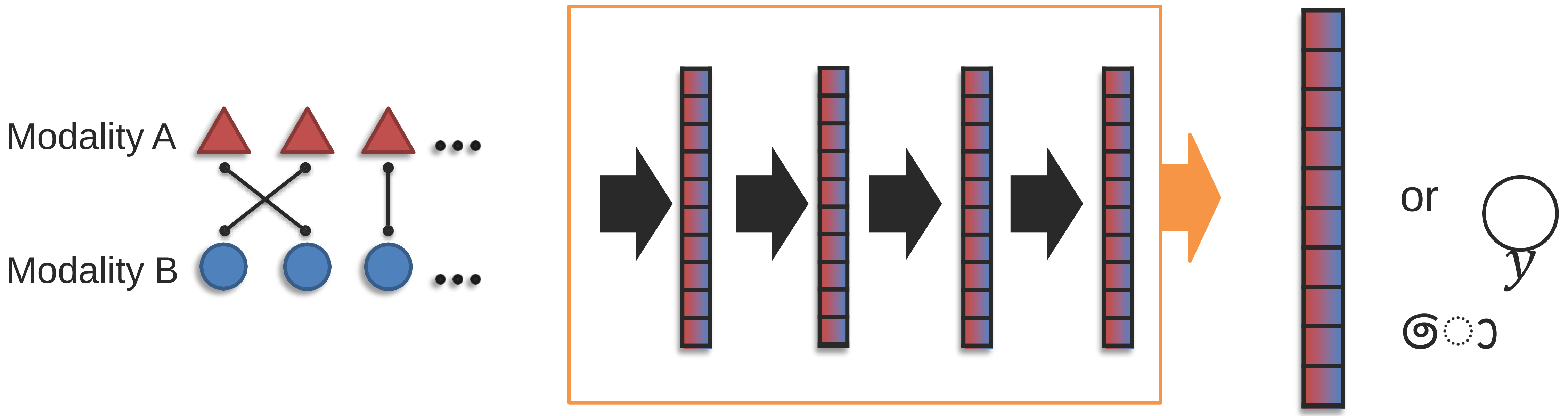
Challenge 2: Alignment

Final Summary

Sub-Challenge	Definition	Concrete Example
Connections (Explicit Alignment)	Direct mapping between modalities	Image captioning (matching objects to words), Speech-to-Text alignment
Aligned Representation	Learning shared multimodal embeddings	CLIP (image-text matching), Lip-reading for speech recognition
Elements (Segmentation)	Breaking multimodal data into meaningful units	Speech phoneme segmentation, Video motion segmentation

Challenge 3: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure to derive meaningful conclusions.



Challenge 3: Reasoning

"Learning to Reason: End-to-End Module Networks for Visual Question Answering"
Andreas et al., NeurIPS 2016

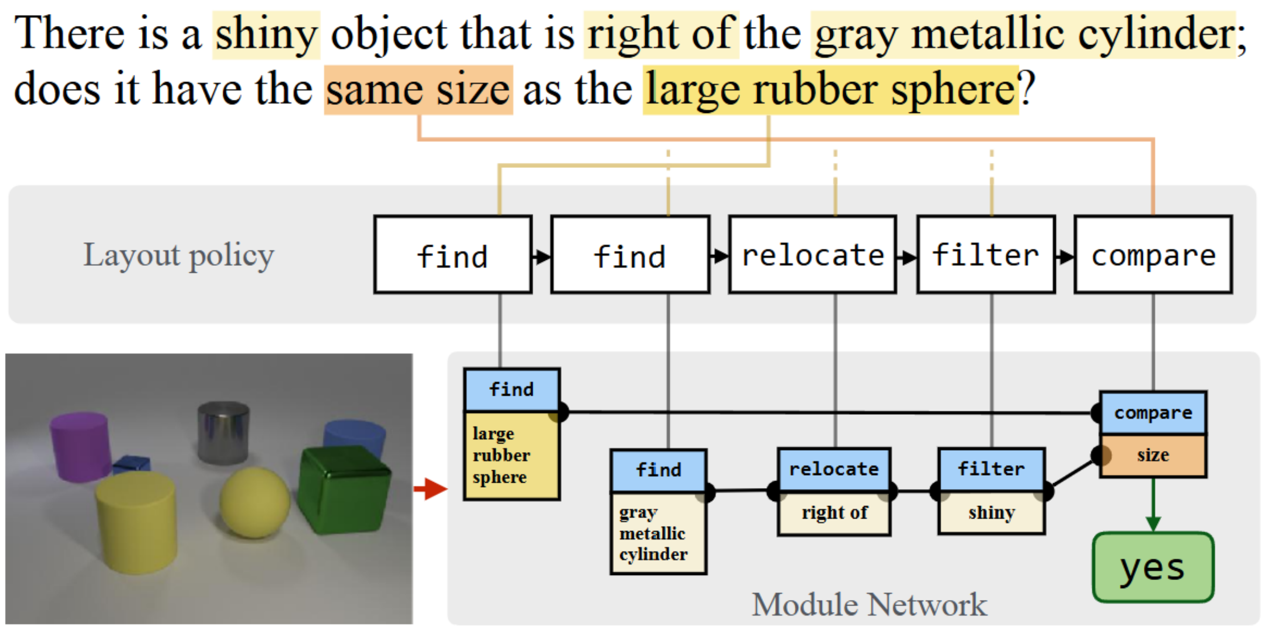
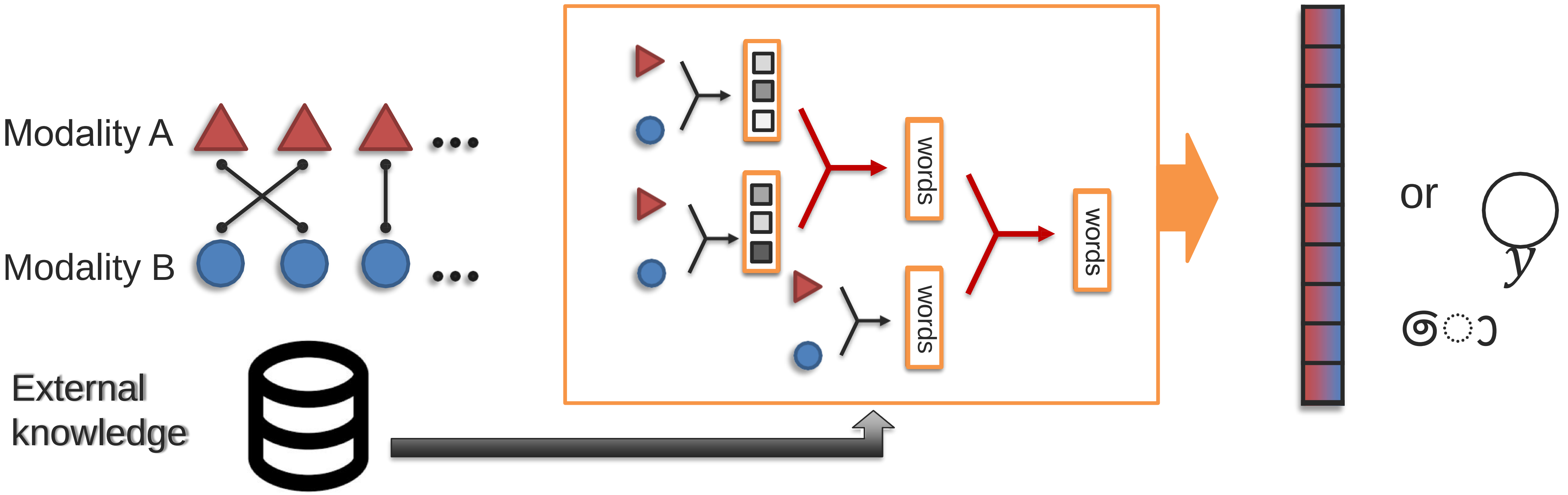


Figure 1: For each instance, our model predicts a computational expression and a sequence of attentive module parameterizations. It uses these to assemble a concrete network architecture, and then executes the assembled neural module network to output an answer for visual question answering. (The example shows a real structure predicted by our model, with text attention maps simplified for clarity.)

Challenge 3: Reasoning

Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure to derive meaningful conclusions.



Challenge 3: Reasoning

Q: Which American president is associated with the stuffed animal seen here?

A: Teddy Roosevelt

"OK-VQA: A Visual Question Answering Benchmark Requiring External Knowledge"

Marino et al., CVPR 2019

A dataset requiring **external commonsense knowledge** for reasoning-based question answering.

Outside Knowledge

Another lasting, popular legacy of Roosevelt is the stuffed toy bears—teddy bears—named after him following an incident on a hunting trip in Mississippi in 1902.

Developed apparently simultaneously by toymakers ... and named after President Theodore "Teddy" Roosevelt, the teddy bear became an iconic children's toy, celebrated in story, song, and film.

At the same time in the USA, Morris Michtom created the first teddy bear, after being inspired by a drawing of Theodore "Teddy" Roosevelt with a bear cub.

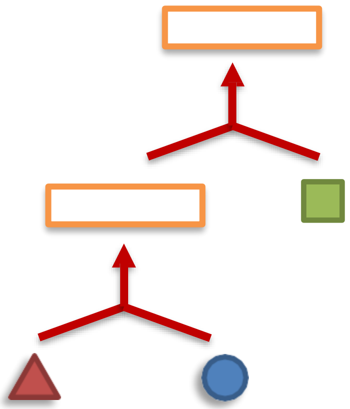
Figure 1: We propose a novel dataset for visual question answering, where the questions require external knowledge resources to be answered. In this example, the visual content of the image is not sufficient to answer the question. A set of facts about teddy bears makes the connection between teddy bear and the American president, which enables answering the question.

Challenge 3: Reasoning

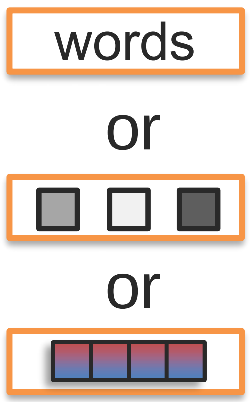
Definition: Combining knowledge, usually through multiple inferential steps, exploiting multimodal alignment and problem structure

Sub-challenges:

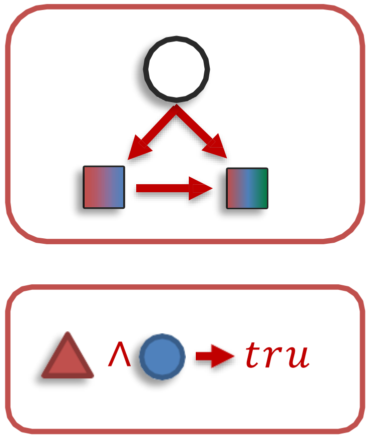
Structure Modeling



Intermediate concepts



Inference Paradigm



External Knowledge

Challenge 3: Reasoning

Structure Modeling

Definition:

•Capturing relationships **within** and **across modalities** by learning structured representations.

•Can be **hierarchical, relational, or graph-based**.

Example 1: Scene Graphs for Visual Reasoning

•**Task**: Understand an image by reasoning about object relationships.

•**Challenge**: Objects in images exist in a structured **spatial** hierarchy.

•**Solution**: Use **scene graphs** to model relationships.

•**Concrete Example**:

- **Image**: "A person sitting on a chair holding a laptop."
- **Scene Graph**:
  - Person → sitting on → Chair
  - Person → holding → Laptop
- **Reasoning**: If "Person is holding Laptop", the **laptop is likely in use**.

Challenge 3: Reasoning

"Neural-Symbolic VQA: Disentangling Reasoning from Vision and Language Understanding" (Zhou et al., NeurIPS 2020)

•Proposes a hybrid **neural-symbolic reasoning** framework for VQA.

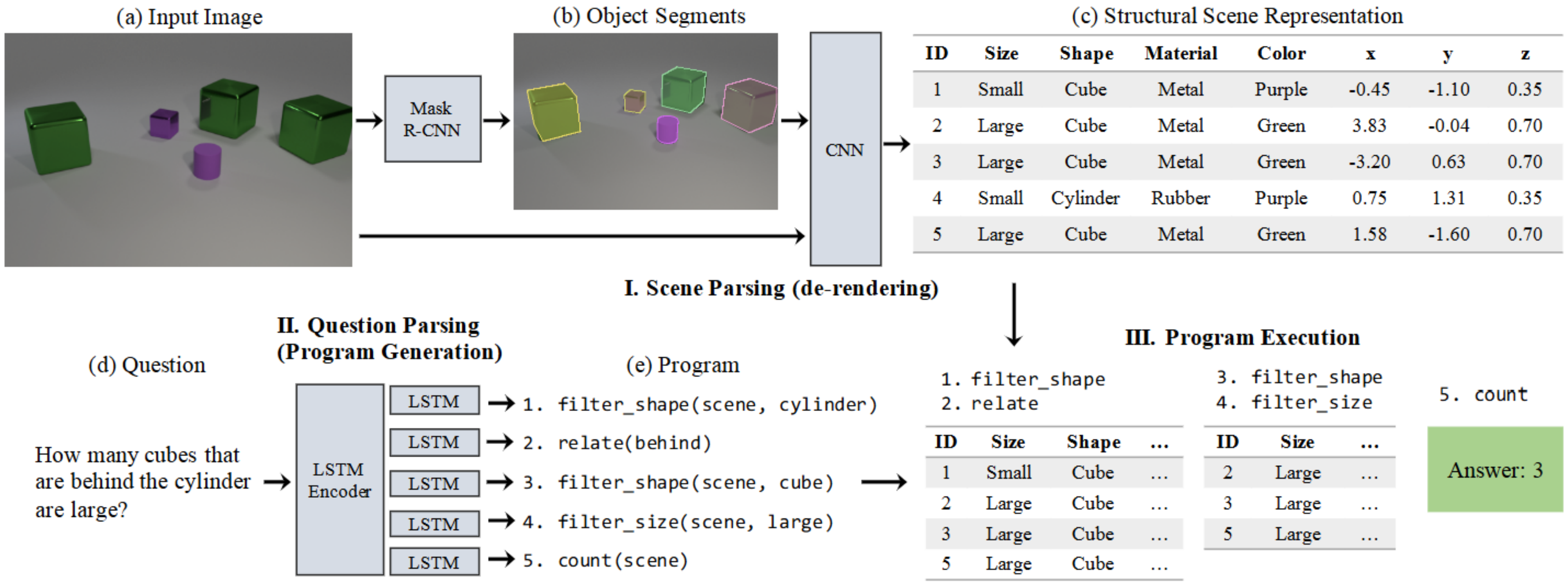


Figure 2: Our model has three components: first, a scene parser (de-renderer) that segments an input image (a-b) and recovers a structural scene representation (c); second, a question parser (program generator) that converts a question in natural language (d) into a program (e); third, a program executor that runs the program on the structural scene representation to obtain the answer.

Challenge 3: Reasoning

Structure Modeling

Definition:

•Capturing relationships **within** and **across modalities** by learning structured representations.

•Can be **hierarchical, relational, or graph-based**.

Example 2: Graph-Based Multimodal Summarization

•**Task**: Summarize news articles with **both text and images**.

•**Solution**: Construct a **multimodal knowledge graph**:

- **Nodes**: Entities (e.g., "Earthquake", "Rescue Teams").
- **Edges**: Relations (e.g., "causes", "affects").

•**Outcome**: Summarization aligns both **textual facts and visual evidence**.

Challenge 3: Reasoning

Intermediate Concepts

Definition:

- Extracting **conceptual representations** between raw data and final inference.
- Helps **bridge the gap** between **modalities**.

Example 1: Multimodal Sentiment Analysis

- **Task**: Detect sentiment from **text, speech, and facial expressions**.
- **Challenge**: Directly mapping raw speech or facial features to emotions is **hard**.
- **Solution**: Use **intermediate concepts**:
    - **Text** → Convert to **word embeddings**.
    - **Speech** → Extract **prosody features** (pitch, energy).
    - **Facial expressions** → Convert to **action units** (e.g., smile intensity).
- **Outcome**: Combine all features to **classify sentiment**.

Challenge 3: Reasoning

Intermediate Concepts

Definition:

•Extracting **conceptual representations** between raw data and final inference.

•Helps **bridge the gap** between **modalities**.

Example 2: Self-Supervised Learning for Audio-Visual Representation

•**Task**: Train a **lip-reading model** without labeled data.

•**Solution**:

- **Intermediate step**: Learn **phoneme representations** before predicting words.
- **Example**:
  - Audio: /th/ /a/ /n/ /k/ /s/
  - Video: **Lip motion sequences**
- **Inference**: Align **audio and visual phonemes** → Predict **words**.

Challenge 3: Reasoning

Inference Paradigm

Definition:

•The logical framework for combining multimodal data to derive conclusions.

•Can be **probabilistic, neural-based, or logical reasoning**.

Example 1: Visual Question Answering (VQA)

•**Task**: Answer a question about an image.

•**Challenge**: Requires **joint reasoning** over text and images.

•**Solution**:

- **Vision Transformer (ViT)** extracts image features.
- **BERT-based model** processes the text question.
- **Multimodal fusion layer** merges them.

•**Concrete Example**:

- **Image**: A red bus in front of a building.
- **Question**: "What color is the bus?"
- **Inference Path**:
  - Detect **"bus" in the image**.
  - Find **color attribute → "red"**.
  - **Final Answer**: "Red."

Challenge 3: Reasoning

Inference Paradigm

Definition:

•The logical framework for combining multimodal data to derive conclusions.

•Can be **probabilistic, neural-based, or logical reasoning**.

Example 2: Logical Reasoning in Multimodal AI

•**Task**: Determine relationships between events in news articles and images.

•**Challenge**: Requires **logical inference**.

•**Solution**: Combine **textual event extraction + image content**.

•**Concrete Example**:

- **Text**: "A hurricane is approaching Florida."
- **Image**: Shows **people evacuating**.
- **Inference**: **If there's an evacuation, the hurricane is severe**.

Challenge 3: Reasoning

External Knowledge

Definition:

•Leveraging **external databases, knowledge graphs, or pretrained models** for better reasoning.

Example 1: Knowledge-Augmented Chatbots

•**Task**: A multimodal AI assistant answering **general knowledge** questions.

•**Challenge**: Image or text alone may **not be enough** for deep understanding.

•**Solution**: Use **external structured knowledge (e.g., Wikipedia, ConceptNet)**.

•**Concrete Example**:

- **Input**: Image of **Tesla Model X**.
- **Question**: "What is the battery range of this car?"
- **AI Process**:
  - Identify car as **Tesla Model X** (Vision).
  - Retrieve battery range from **Wikipedia API**.
  - Generate response: **"It has a range of ~350 miles per charge."**

Challenge 3: Reasoning

External Knowledge

Definition:

•Leveraging **external databases, knowledge graphs, or pretrained models** for better reasoning.

Example 2: Commonsense Reasoning with Knowledge Graphs

•**Task**: Predict **cause-effect relationships** in multimodal data.

•**Challenge**: Need **real-world commonsense knowledge**.

•**Solution**: Use **ConceptNet or COMET (Commonsense Transformer)**.

•**Concrete Example**:

- **Text**: "John forgot his umbrella."
- **Image**: Shows **John standing in the rain, wet**.
- **External Knowledge**:
  - **ConceptNet Fact**: "If you don't carry an umbrella, you get wet in rain."
- **Inference**: **"John is wet because he forgot his umbrella."**

Challenge 3: Reasoning

Final Summary

Sub-Challenge	Definition	Concrete Example
Structure Modeling	Learning structured relationships	Scene graphs for VQA, Graph-based summarization
Intermediate Concepts	Extracting feature representations	Sentiment analysis, Lip-reading phonemes
Inference Paradigm	Logical or probabilistic reasoning over modalities	VQA (Vision-Text fusion), Commonsense AI
External Knowledge	Using external databases or knowledge graphs	Wikipedia-augmented Chatbots, COMET-based causal reasoning

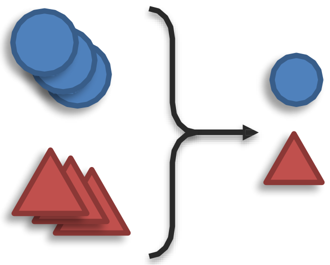
Challenge 4: Generation

Definition: **Learning a generative process** to produce new data while maintaining **cross-modal interactions, structure, and coherence**.

Covers tasks such as **summarization, translation, and creative generation** across multiple modalities.

Sub-challenges:

Summarization



Reduction

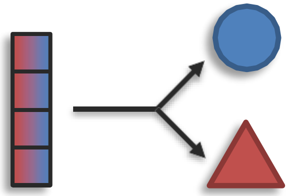
□ > □

Translation

Maintenance

□ = □

Creation



Expansion

□ < □

Information:
(content)

Challenge 4: Generation

Summarization (Reduction of Information)

Definition:

Extracting the most relevant content from multimodal inputs while **reducing redundancy**.

•Important for tasks where raw multimodal data is **too large to process directly**.

Example 1: Video Summarization

•**Task**: Summarize a **long video into a short clip**.

•**Challenge**: The model must **retain key events** while **removing redundant frames**.

•**Solution**:

- Use **Temporal Attention Networks (e.g., TimeSformer)** to extract keyframes.
- Select **the most informative frames** based on visual importance.

•**Concrete Example**:

- Input: **A 30-minute surveillance video**.
- Output: **A 1-minute summary showing all significant actions (e.g., a person entering a restricted area).**

Challenge 4: Generation

Retrospective Encodersfor Video Summarization
Zhang 2018

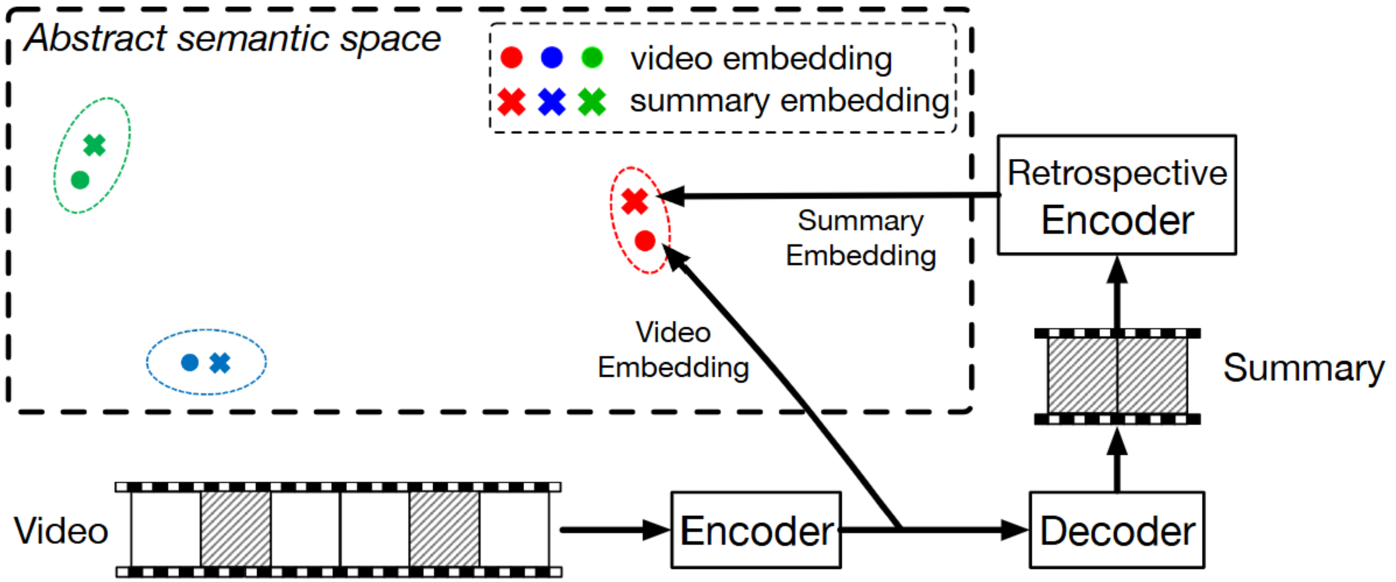


Fig. 1. Conceptual diagram of our approach. Our model consists of two components. First, we use SEQ2SEQ models to generate summaries. Secondly, we introduce an retrospective encoder that maps the generated summaries to an abstract semantic space so that we can measure how well the summaries preserve the information in the original videos, *i.e.* the summary (×) should be close to its original video (•) while distant from other videos ({•, •}) and summaries ({×, ×}). In this semantic space, we derive metric learning based loss functions and combine them with the discriminative loss by matching human generated summaries and predicted ones. See text for details.

Challenge 4: Generation

Summarization (Reduction of Information)

Definition:

Extracting the most relevant content from multimodal inputs while **reducing redundancy**.

•Important for tasks where raw multimodal data is **too large to process directly**.

Example 2: Multimodal News Summarization

•**Task**: Summarize a news article while incorporating **relevant images**.

•**Challenge**: The model must decide **which sentences to retain** and **which images are relevant**.

•**Solution**:

- Use **multimodal transformers (e.g., MM-T5, PEGASUS)**.
- Cross-attend between **text and images**.

•**Concrete Example**:

- Input: **A news article with 5 images**.
- Output: **A 100-word summary with 1-2 key images**.

Challenge 4: Generation

Translation (Cross-Modal Transformation While Preserving Meaning)

Definition:

Translating information from one modality to another while **preserving its original meaning**.

•Unlike summarization, the **amount of content remains the same**, but the **representation changes**.

Example 1: Speech-to-Text Translation (Automatic Speech Recognition)

•**Task**: Convert spoken audio into **text transcripts**.

•**Challenge**: The model must handle **different accents, noise, and speech patterns**.

•**Solution**:

- **Whisper AI** (by OpenAI) or **DeepSpeech** converts speech to text.
- **Alignment models** match **words with timestamps**.

•**Concrete Example**:

- Input: 🎤 "**Hello, how are you today?**" (spoken audio)
- Output: 📝 "**Hello, how are you today?**" (text transcription)

Challenge 4: Generation

Translation (Cross-Modal Transformation While Preserving Meaning)
Definition:
Translating information from one modality to another while **preserving its original meaning**.
•Unlike summarization, the **amount of content remains the same**, but the **representation changes**.
Example 2: Image-to-Text Translation (Image Captioning)
•**Task**: Describe an image using **natural language**.
•**Challenge**: Model must detect **objects, actions, and relationships** in an image.
•**Solution**:
- **Vision-Language Models (e.g., BLIP, LLaVA, Flamingo)** process **both images and text**.

•**Concrete Example**:
- Input: 📷 **Image of a dog playing in a park**.
- Output: 📝 **"A golden retriever is running on the grass."**

Challenge 4: Generation

Show and Tell: A Neural Image Caption Generator, Vinyals 2015

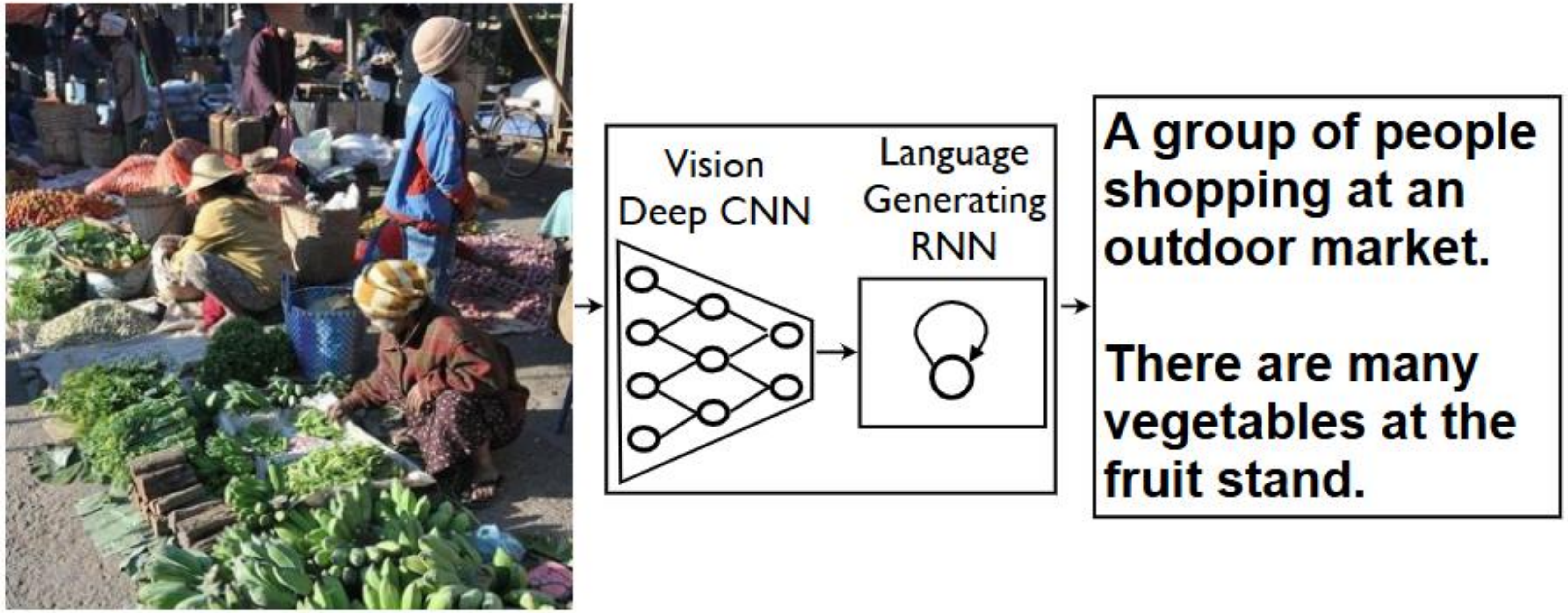


Figure 1. NIC, our model, is based end-to-end on a neural network consisting of a vision CNN followed by a language generating RNN. It generates complete sentences in natural language from an input image, as shown on the example above.

Challenge 4: Generation

Creation (Expansion of Information)

Definition:

Generating new content beyond the input data while **expanding creative possibilities**.

•Used in **text-to-image, text-to-music, and story generation**.

Example 1: Text-to-Image Generation (DALL-E, Stable Diffusion)

•**Task**: Generate images from textual descriptions.

•**Challenge**: The model must ensure **coherence and realism**.

•**Solution**:

- **Diffusion models** refine **noisy images** to generate **high-quality visuals**.

•**Concrete Example**:

- Input: 📝 "**A futuristic city with flying cars and neon lights.**"
- Output: 🎨 **Generated image of a cyberpunk city**.

Challenge 4: Generation

Creation (Expansion of Information)

Definition:

Generating new content beyond the input data while **expanding creative possibilities**.

•Used in **text-to-image, text-to-music, and story generation**.

Example 2: Multimodal Story Generation

•**Task**: Generate an **interactive story with text and images**.

•**Challenge**: The generated images must align with the **story plot**.

•**Solution**:

- **Multimodal GPT (e.g., GPT-4V, Gemini)** generates both **narratives and visuals**.

•**Concrete Example**:

- Input: 📝 "**A knight embarks on a quest to slay a dragon.**"
- Output: 📖 **Text Story + AI-Generated Illustrations.**

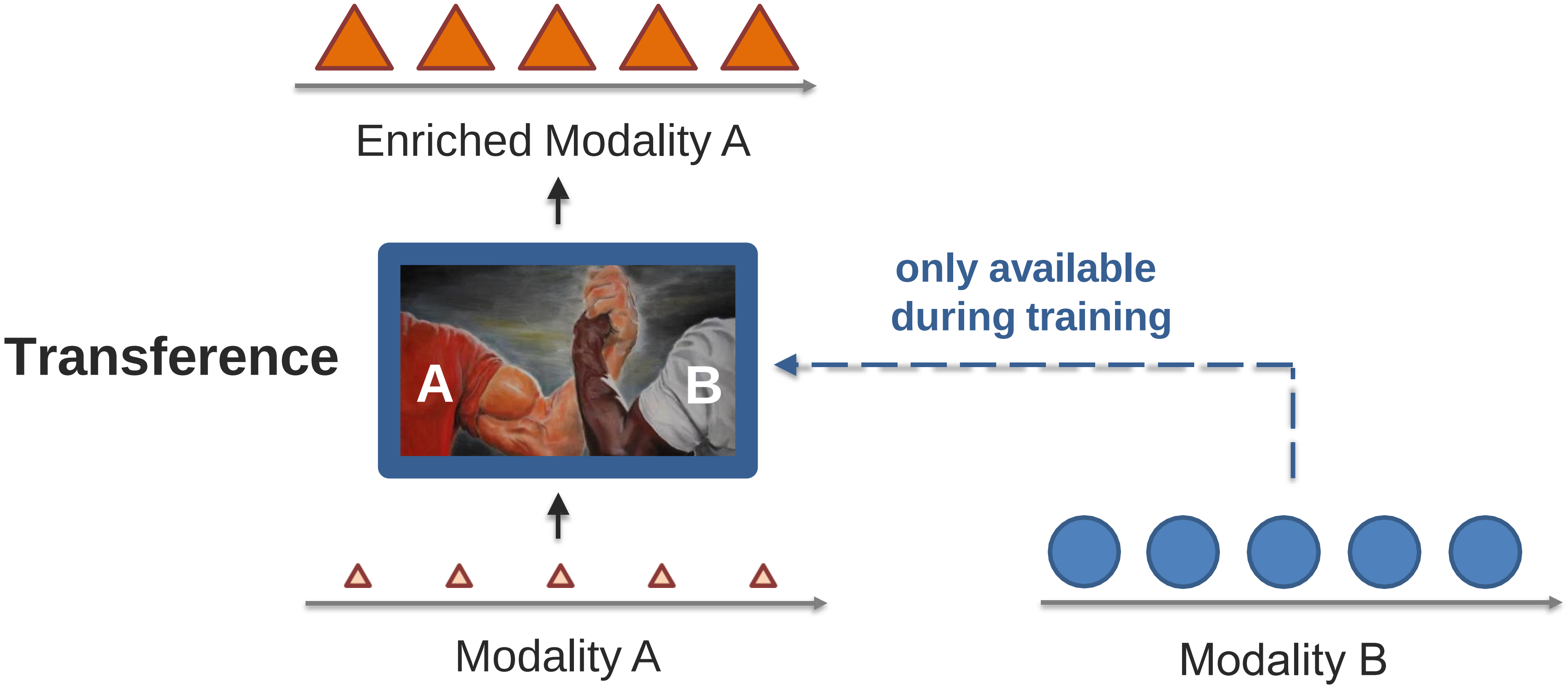
Challenge 4: Generation

Final Summary

Sub-Challenge	Definition	Concrete Example
Summarization (Reduction)	Extract key information while removing redundancy	Video summarization, News article condensation
Translation (Preserving Meaning)	Convert data between modalities while retaining meaning	Speech-to-text, Image captioning
Creation (Expansion)	Generate new content beyond the given data	Text-to-image (DALL-E), Story generation

Challenge 5: Transference

Definition: Transferring knowledge between different modalities to help a **weaker modality** that might be **noisy, incomplete, or resource-limited**. Used in **low-resource learning scenarios**, **data-efficient AI**, and **multimodal adaptation**.



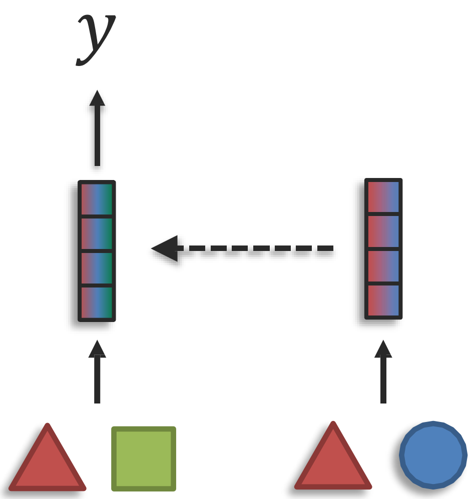
Challenge 5: Transference

Definition: Transferring knowledge between different modalities to help a **weaker modality** that might be **noisy, incomplete, or resource-limited**.

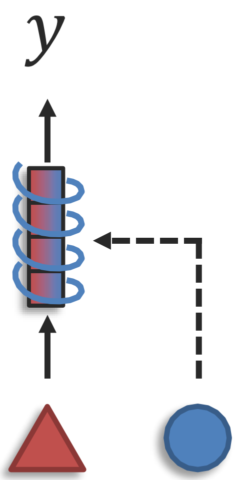
Used in **low-resource learning scenarios**, **data-efficient AI**, and **multimodal adaptation**.

Sub-challenges:

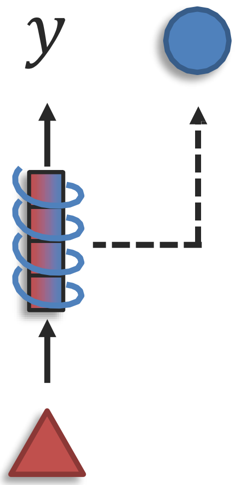
Transfer



Co-learning
via representation



Co-learning
via generation



Challenge 5: Transference

Transfer (Cross-Modal Knowledge Transfer)

Definition:

•One modality helps another learn better representations or predictions.

•Applied when one **modality lacks sufficient labeled data** or is **less informative alone**.

Example 1: Zero-Shot Learning for Speech Recognition (Text Helps Speech)

•**Task**: Train a speech recognition model in a **low-resource language (e.g., Swahili)**.

•**Challenge**: The **Swahili speech dataset is small**, but a **large text corpus exists**.

•**Solution**:

- Train a **text-based language model (BERT)** on Swahili text.
- Transfer learned word relationships to the **speech model** (wav2vec2).

•**Concrete Example**:

- Swahili has **limited speech data** but **extensive text data**.
- A model **pretrained on text** is **fine-tuned on speech**, improving accuracy.

Challenge 5: Transference

Transfer (Cross-Modal Knowledge Transfer)

Definition:

•One modality helps another learn better representations or predictions.

•Applied when one **modality lacks sufficient labeled data** or is **less informative alone**.

Example 2: Visual-Language Pretraining for Medical Diagnosis

•**Task**: Improve **X-ray classification using radiology reports**.

•**Challenge**: Medical images are **hard to interpret without expert labeling**.

•**Solution**:

- Train a **BERT-like model on medical text**.
- Transfer knowledge to an **image-based model (ResNet or ViT)**.

•**Concrete Example**:

- Input: **X-ray scan**.
- Output: **"Pneumonia detected"**, aided by text-based diagnosis.

Challenge 5: Transference

Co-Learning via Representation

Definition:

•Learning **joint representations across modalities** so they benefit from each other.

•**Improves robustness** when one modality is **missing or weak**.

Example 1: Audio-Visual Speech Recognition

•**Task**: Improve speech recognition by using **both audio and lip movements**.

•**Challenge**: Noisy environments degrade **audio**, making **speech harder to recognize**.

•**Solution**:

- Train a **multimodal encoder** that aligns **lip movements (video) and speech (audio)**.
- If **audio is unclear**, the system **relies more on lip motion**.

•**Concrete Example**:

- **In a noisy subway**, the **lip-reading model corrects errors in speech-to-text output**.

Challenge 5: Transference

Co-Learning via Representation

Definition:

•Learning **joint representations across modalities** so they benefit from each other.

•**Improves robustness** when one modality is **missing or weak**.

Example 2: Sensor Fusion for Autonomous Vehicles

•**Task**: Improve self-driving car perception by combining **LiDAR and camera data**.

•**Challenge**: Cameras struggle in **low light**, but LiDAR works in all conditions.

•**Solution**:

- Train a **joint model** where **camera feeds and LiDAR point clouds** share a representation.
- **When visibility is poor**, the model **relies more on LiDAR**.

•**Concrete Example**:

- **A self-driving car detects pedestrians at night** using **LiDAR when cameras fail**.

Challenge 5: Transference

Co-Learning via Generation

Definition:

•One modality **helps generate missing or complementary data** for another.

•Often used in **data augmentation and multimodal simulation**.

Example 1: Text-to-Speech Synthesis (TTS)

•**Task**: Generate **natural speech** from **text input**.

•**Challenge**: Speech synthesis needs **expressive and realistic prosody**.

•**Solution**:

- A **text encoder (GPT-3)** generates phoneme sequences.
- A **speech decoder (Tacotron, WaveNet)** generates **audio waveforms**.

•**Concrete Example**:

- Input: **Text: "Welcome to the AI conference!"**.
- Output: **Synthetic voice that sounds natural**.

Challenge 5: Transference

Co-Learning via Generation

Definition:

•One modality **helps generate missing or complementary data** for another.

•Often used in **data augmentation and multimodal simulation**.

Example 2: Image-to-Text Augmentation for Captioning Models

•**Task**: Train an **image captioning model** when **text annotations are scarce**.

•**Challenge**: The dataset has **millions of images but few captions**.

•**Solution**:

- Use **DALL-E to generate synthetic images** based on descriptions.
- Train the captioning model on **synthetic + real data**.

•**Concrete Example**:

- **DALL-E generates missing training samples** for **low-resource image datasets**.

Challenge 5: Transference

Final Summary

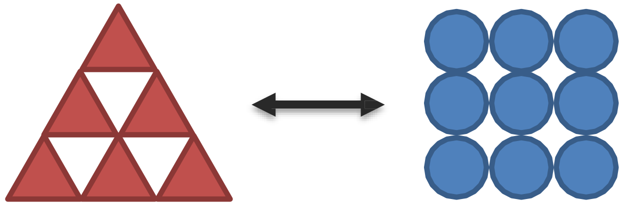
Sub-Challenge	Definition	Concrete Example
Transfer	One modality transfers knowledge to another	Text-trained models help low-resource speech recognition
Co-Learning via Representation	Learning shared representations for robustness	Lip-reading assists noisy speech recognition
Co-Learning via Generation	One modality generates missing data for another	Text-to-speech, synthetic image-text augmentation

Challenge 6: Quantification

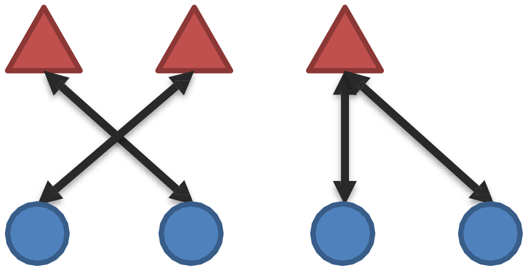
Definition: **Empirical and theoretical study** to improve understanding of **heterogeneity, cross-modal interactions, and learning dynamics** in multimodal AI.
Involves **metrics, evaluation techniques, and optimization strategies** to enhance multimodal learning.

Sub-challenges:

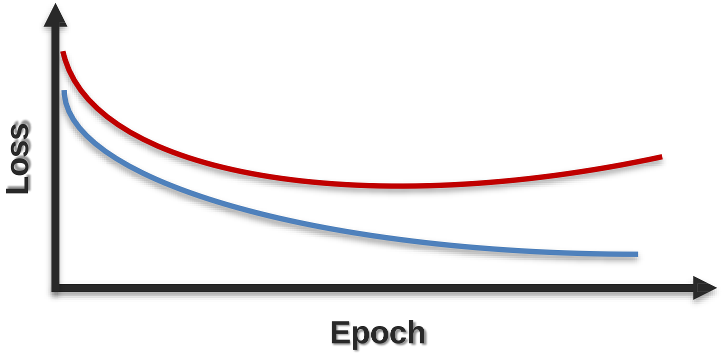
Heterogeneity



Interactions



Learning



Challenge 6: Quantification

Heterogeneity (Quantifying Differences Across Modalities)

Definition:

•**Different modalities have diverse structures, scales, and distributions**, making quantification complex.

•Measuring **how much information is unique vs. redundant** across modalities.

Example 1: Measuring Modality Contribution in Multimodal Learning

•**Task**: Evaluate the importance of **each modality in a multimodal model**.

•**Challenge**: Some modalities might dominate, reducing **effective fusion**.

•**Solution**:

- Compute **Shannon entropy** for **text, images, and speech** separately.
- Use **Ablation Studies**: Train models with and without certain modalities to measure performance drop.

•**Concrete Example**:

- Model trained on **text + image** vs. **text-only**.
- If performance drops **only slightly**, the image modality is **less informative**.

Challenge 6: Quantification

Heterogeneity (Quantifying Differences Across Modalities)

Definition:

•**Different modalities have diverse structures, scales, and distributions**, making quantification complex.

•Measuring **how much information is unique vs. redundant** across modalities.

Example 2: Quantifying Inconsistencies in Multimodal Data

•**Task**: Detect **disagreements** between modalities in AI decision-making.

•**Challenge**: **Speech and facial expressions** might convey **different emotions**.

•**Solution**:

- Compute **KL divergence** between probability distributions of each modality.

•**Concrete Example**:

- **Audio predicts "happy"** (80% confidence).
- **Face predicts "neutral"** (75% confidence).
- **Divergence score quantifies inconsistency**.

Challenge 6: Quantification

Interactions (Measuring Cross-Modal Dependencies)
Definition:
•**Quantifying how strongly modalities influence each other**.
•Ensuring that interactions are **meaningful rather than redundant**.
Example 1: Measuring Cross-Modal Attention in Transformers
•**Task**: Evaluate how much **text influences image embeddings** in CLIP models.
•**Challenge**: Text and images may not align **perfectly**, leading to **misinterpretation**.
•**Solution**:
- Compute **attention weight distributions** from **vision-language transformers (ViLBERT, CLIP)**.

•**Concrete Example**:
- **Text prompt: "A dog in the park."**
- **Heatmap of attention weights** shows **which image regions are linked to words**.
- High alignment = **strong interaction**.

Challenge 6: Quantification

Interactions (Measuring Cross-Modal Dependencies)

Definition:

•**Quantifying how strongly modalities influence each other**.

•Ensuring that interactions are **meaningful rather than redundant**.

Example 2: Quantifying Temporal Synchronization in Audio-Visual Models

•**Task**: Measure **how well speech and lip movements align** in videos.

•**Challenge**: Some words are **spoken faster** than their corresponding lip movements.

•**Solution**:

- Compute **Dynamic Time Warping (DTW)** distance between speech and lip-motion embeddings.

•**Concrete Example**:

- **Low DTW score** = Good synchronization.
- **High DTW score** = Misalignment (e.g., dubbed movies).

Challenge 6: Quantification

Learning (Evaluating Multimodal Learning Dynamics)

Definition:

- **Studying how multimodal models converge**, generalize, and optimize.
- Ensuring balanced learning across modalities.

Example 1: Measuring Learning Speed Across Modalities

- **Task**: Compare the **convergence rate** of text-only, image-only, and multimodal models.
- **Challenge**: Some modalities **learn faster than others**, causing **imbalanced gradients**.
- **Solution**:
  - Track **loss curves** for each modality separately.
- **Concrete Example**:
  - **Text model loss drops fast**, while **image model learns slower**.
  - Adaptive learning rates adjust the **modality learning pace**.

Challenge 6: Quantification

Learning (Evaluating Multimodal Learning Dynamics)
Definition:
•**Studying how multimodal models converge**, generalize, and optimize.
•Ensuring balanced learning across modalities.
Example 2: Evaluating Catastrophic Forgetting in Multimodal Models
•**Task**: Assess whether a multimodal model **forgets** older learned modalities over time.
•**Challenge**: **When fine-tuning** on one modality, other modalities might be **forgotten**.
•**Solution**:
- Measure **accuracy degradation** on past tasks before and after fine-tuning.

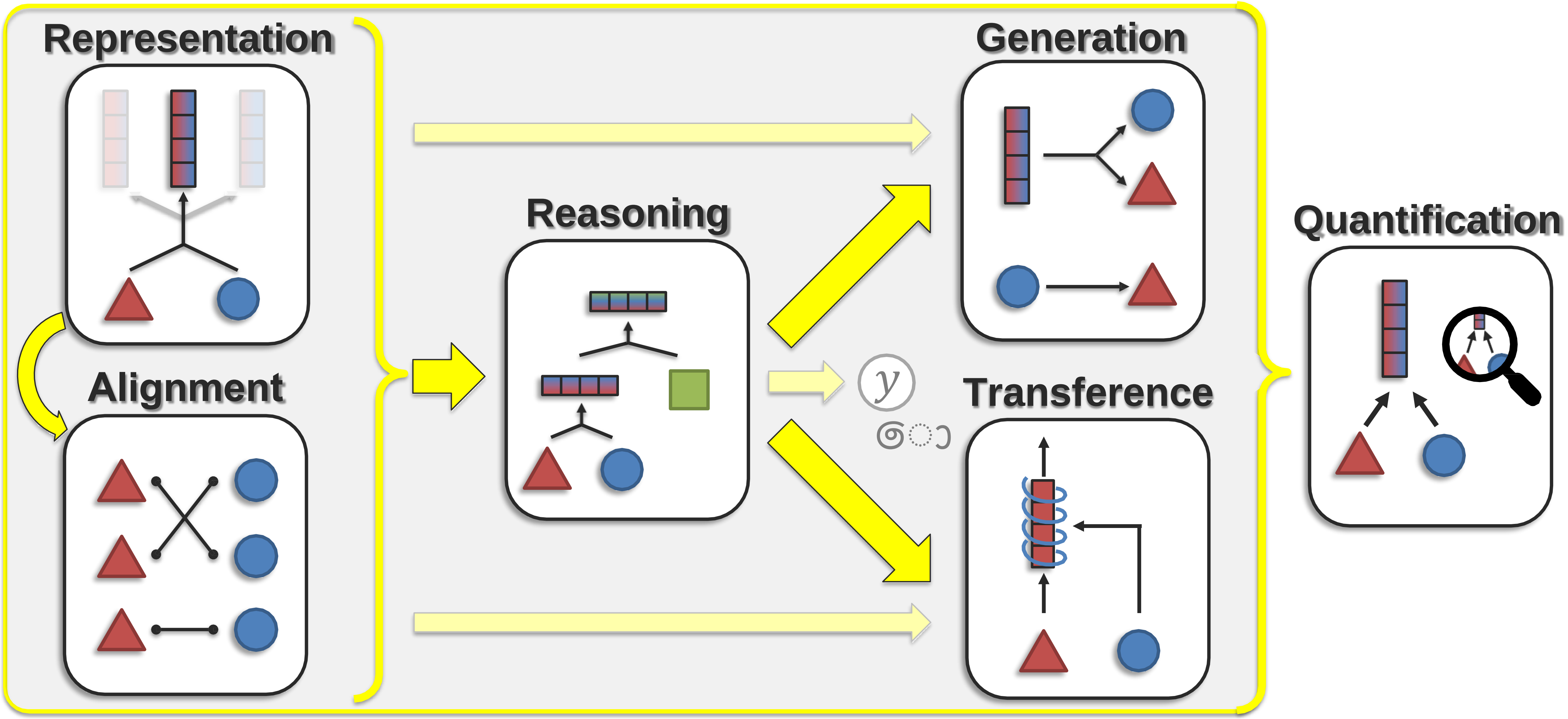
•**Concrete Example**:
- **Image-text model fine-tuned on new images**.
- Text understanding **accuracy drops by 10%**, indicating **modality forgetting**.

Challenge 6: Quantification

Final Summary

Sub-Challenge	Definition	Concrete Example
Heterogeneity	Quantifying differences across modalities	Measuring **modality importance** in model performance
Interactions	Evaluating cross-modal dependencies	Attention weights in vision-language transformers
Learning	Studying optimization and generalization	Tracking **loss convergence for each modality**

Core Multimodal Challenges



Multimodal Machine Learning Models

Models for Multimodal Learning

1. **Concatenation-based Models** (e.g., Early Fusion)
2. **Attention-based Models** (e.g., Cross-modal transformers like CLIP, ViLBERT)
3. **Graph Neural Networks (GNNs)** (e.g., Graph-based multimodal learning)
4. **Contrastive Learning** (e.g., Aligning image-text embeddings)

Future of Multimodal Learning

Research Challenges

1. **Multimodal Explainability**
   1. Understanding how models integrate different modalities.
2. **Efficient & Scalable Fusion**
   1. Reducing computation for real-time multimodal applications.
3. **Few-Shot & Zero-Shot Learning**
   1. Generalizing to unseen modalities without retraining.

Emerging Trends

- **GPT-4V** (multimodal ChatGPT).
- **Self-Supervised Multimodal Learning** (reducing reliance on labeled data).
- **Embodied AI** (robots using multimodal perception for interaction).